

The Role of the Dopamine D₁ Receptor in the Behavioural and Reinforcing Properties of Cocaine and Methamphetamine

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Abstract

The current study examined the role of the dopamine D₁ receptor in the behavioural and reinforcing properties of cocaine and methamphetamine. Using a mutant rat model, we carried out detailed locomotor analysis, conditioned taste avoidance, and conditioned place preference experiments. Of particular interest to this study was whether the role of the D₁ receptor is dissociable for cocaine and methamphetamine. These drugs are both classified as psychostimulants and exert their effects via the dopamine transporter. However, research has shown that the specific binding profiles of these drugs, as well as other mechanisms by which they work are different. Major findings indicate that the D₁ receptor is more involved in the behavioural and reinforcing processes associated with methamphetamine than cocaine. This research also provides further insight into the D₁ mutant rat model.

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The Role of the Dopamine D₁ Receptor in the Behavioural and Reinforcing Properties of Cocaine and Methamphetamine

Drug dependence is a chronic and relapsing disorder that negatively impacts both sufferers and the wider community. In the United States alone it is estimated that 67 billion dollars is spent annually on matters related to drug addiction (McLellan, Lewis, O'Brien, & Kleber, 2000). A longitudinal study carried out within New Zealand followed over 1,200 children until the age of 25 and found that over 75% of participants had sampled illicit substances. While the majority of drug users in this population did not go on to develop dependency, about 15% did show symptoms associated with illicit drug dependence, indicating that drug use and abuse is a problem affecting the New Zealand community (Boden, Fergusson, & John Horwood, 2006). While small advances are constantly made in the understanding and treatment of drug dependence, one problem that continues to plague this area of research is the individual variability within drug users in regards to susceptibility to dependence, and response to treatment programs. For this reason, it is imperative that research is carried out in order to deepen our understanding of the molecular and genetic underpinnings of drug use and abuse. Findings in this area can then be applied to the development of effective treatment programs, which can lead to individual and societal harm minimisation in the area of drug dependence.

Dopamine (dihydroxyphenylethylamine) is a very important neurotransmitter that is widely distributed within the central nervous system (Zhang, Xiong, Zhen, & Zhang, 2009). Dopamine is involved in the regulation of many fundamental neurobehavioral functions, including learning, cognition, motivation, movement and reward (Undieh, 2010). The involvement of the dopaminergic system in so many functions also means that faulty dopaminergic signalling can lead to the development of many different psychiatric disorders, including schizophrenia, depression and drug addiction (Di Chiara, 1995)

Dopamine interacts with five different G-protein coupled receptors that are divided into two separate receptor subtype families. There is the D₁-like family that includes both D₁ and D₅ receptors (or equivalently, D_{1A} and D_{1B} receptors in rodents), and the D₂-like family which is made up of D₂, D₃, and D₄ receptors (Nicola, Surmeier, & Malenka, 2000). The different families of receptors are classified as such due to their influence on adenylyl cyclase. The D₁-like family activates G proteins that stimulate adenylyl cyclase whereas, the D₂ family of receptors inhibit the formation of adenylyl cyclase due to being coupled to G_i, and also interacting with other second messenger proteins (Missale, Nash, Robinson, Jaber, & Caron, 1998; Undieh, 2010; Zhang et al., 2009). Of all dopamine receptor subtypes the D₁ receptor is the most widely expressed (Zhang et al., 2009). It is abundantly found in the striatum, cerebral cortex and olfactory bulb, and also expressed on a smaller scale within the hippocampus and amygdala (Undieh, 2010). Although part of the same family, D₁ and D₅ receptors are differentially localized throughout brain tissue and show different patterns of subcellular distribution. Compared to D₁, D₅ receptors examined in rat and monkey brains exhibited a more widespread distribution across neural regions such as the amygdala, hippocampus, hypothalamus, thalamus, and cerebellum. Specifically, D₅ receptor distribution was found to be largely dissociable to D₁ receptor expression within subcortical brain structures (Bergson et al., 1995; Ciliax et al., 2000; Undieh, 2010). Of interest to the current study is the role of the D₁ receptor in reward, specifically the rewarding properties of methamphetamine and cocaine. Research identifying the distinct roles of receptors is important to advance current knowledge surrounding different brain processes and brain disorders. If we can tease apart which specific receptors are involved in different behaviours, then we will be better equipped to develop target-based pharmacotherapies.

A wealth of literature has implicated the mesocorticolimbic dopamine system, as a key contributor to the reinforcing properties of many drugs of abuse. This knowledge stems

from early research conducted by Olds and Milner (1954) who examined the reinforcing effects of electrode stimulation in certain regions of the rat brain. This work indicated that there are specific areas in the brain where rats will readily stimulate themselves, suggesting that these regions are involved in reward. Subsequent research expanded on these findings and illustrated that the midbrain dopamine system was particularly sensitive to electrical self-stimulation. Research examining the consequences of dopamine lesioning and pharmacological manipulation eventually lead to the hypothesis that mesolimbic dopaminergic neurons act as a final common pathway, mediating reinforcement processes within the central nervous system (Spanagel & Weiss, 1999). The mesolimbic dopamine system is made up of dopaminergic neurons that project from the ventral tegmental area to the nucleus accumbens shell region. Both naturally occurring rewarding stimuli, as well as drugs of abuse, have been shown to increase extracellular levels of dopamine in the nucleus accumbens (Koob, 2000). The idea that mesolimbic dopamine acts as a final common pathway for reinforcement has now been prominent throughout the literature for many years (Pierce & Kumaresan, 2006). In addition to this, the shift from voluntary controlled drug use, to chronic relapsing and compulsive drug taking behaviour (a defining feature of drug dependence) is believed to be related to a shift in striatal activation (Everitt & Robbins, 2013). It has been found that in comparison with acute drug administration, well-established cocaine seeking behaviour in rats leads to increased dopamine release in the anterior dorsolateral striatum, but not the nucleus accumbens core or shell region (Ito, Dalley, Robbins, & Everitt, 2002).

While there is a large amount of evidence that dopamine is involved in general reward, it is less clear which specific processes it is involved in, as well as which dopamine receptor subtypes are most important. The current study is interested in identifying the specific role of the dopamine D₁ receptor in the rewarding processes of drugs of abuse.

Specifically, the role the D₁ receptor plays in the reinforcing properties of cocaine and methamphetamine. It is believed that dopamine receptors are involved in mediating two critical aspects of addiction; primary reward and reward seeking behaviour (Self, Belluzzi, Kossuth, & Stein, 1996). When reviewing the previous literature surrounding methamphetamine and reward, studies using *d*-amphetamine will also be discussed when research using methamphetamine is lacking. These substances are very similar, and it has been shown that the pharmacological profile of both agents in respect to monoamine release and uptake inhibition are largely comparable (Rothman & Baumann, 2003).

Self-administration is believed to be the most analogous animal model of human drug taking behaviour (Panlilio & Goldberg, 2007). In a typical self-administration paradigm an animal is trained to perform an operant response such as a lever press, nose poke, or similar, in order to receive an intravenous (IV) infusion of drug. Drugs that are readily self-administered by animals appear to correlate with drugs that are highly used and abused in humans (Haney, Collins, Ward, Foltin, & Fischman, 1999). Research using pharmacological agents has found that dopamine agonists can be used to support self-administration behaviour. This illustrates that dopamine receptors are important for mediating the primary reward of a stimulus. This idea was shown by Woolverton, Goldberg, and Ginos (1984) who were able to train 21 female rats to self-administer apomorphine. Apomorphine is a non-selective dopamine agonist, which works by stimulating both D₁-like and D₂-like receptors. Once the animals had acquired apomorphine self-administration, they were pre-treated with pimozide, which resulted in a disruption of self-administration behaviour. Pimozide, in contrast to apomorphine, is a dopamine antagonist with a particularly high affinity for the D₂ receptor (although it still works on several other receptor subtypes including the D₁ receptor). The findings of this early study illustrate that activation of dopamine receptors act as a reinforcer to establish and maintain self-administration behaviour, much like drugs of abuse.

However, when dopamine receptor stimulation is blocked, the reinforcing effects are abolished thereby decreasing self-administration behaviour.

While these findings clearly implicate dopamine in reinforcement, subsequent research looking into which specific receptor subtypes are important in this process was still needed. Original findings in this area seemed to suggest that the D₂-like receptors are more involved in reinforcement processes than the D₁-like family (Self et al., 1996). Research carried out by Woolverton et al. (1984) examined IV self-administration of dopamine agonists in rhesus monkeys. Animals were first trained to lever press for an IV infusion of either cocaine or *d*-amphetamine. Following this, a number of different dopamine receptor agonists were substituted for the baseline drug in order to investigate which pharmacological agents would maintain responding. The study examined apomorphine, piribedil (a D₂/D₃ receptor agonist), bromocriptine (a D₂ agonist), propylbutyldopamine (PBDA – a dopamine agonist that works on both D₁ and D₂ receptors), and SKF 38393 (a selective D₁/D₅ partial agonist). Findings indicated that successful self-administration substitution was limited to the agents that stimulated the D₂ receptor (apomorphine, bromocriptine, PBDA and piribedil). SKF 38393, the only more or less selective D₁-like receptor (partial) agonist that was tested was not successful at maintaining self-administration behaviour when substituted in for either cocaine or *d*-amphetamine. These results can be interpreted to mean that it is specifically dopamine agonists targeting the D₂ receptor that have the ability to function as a positive reinforcer, whereas the D₁ receptor subtype is not as important in drug reinforcement processes. Although, the fact that SKF38393 is only a partial agonist may also have contributed to its lack of effect.

Subsequent studies have found that many different species of animals will self-administer D₁-like agonists with higher efficacy, such as SKF 81297 and SKF 82958, which are both D₁/D₅ receptor full agonists (Self, 2010). Grech, Spealman, and Bergman (1996)

examined D₁ agonist self-administration in squirrel monkeys. Three different pharmacological agents with differing D₁ receptor selectivity and efficacy (SKF 81297, SKF 82958, and SKF 77434) were administered to animals that had previously been trained to self-administer cocaine. Cocaine and D₁ agonist self-administration was tested under both fixed ratio (FR) and second order fixed interval (FI) schedules of IV injection. Findings illustrated that SKF 82958 had the ability to maintain responding on both FR and FI schedules of reinforcement, whereas SKF 81297 only maintained responding under FR conditions and SKF 77434 failed to sustain self-administration behaviour on either schedule. These findings indicate that the selectivity and efficacy of such pharmacological agents affects the ability of these drugs to act as substitutes for cocaine in a self-administration paradigm.

Research has also shown that laboratory rats will self-administer D₁ dopamine receptor agonists. Following the previously discussed finding that the D₁ partial agonist SKF 38393 was not self-administered by rodents, Self and Stein (1992) investigated this further by testing two derivatives of this drug - SKF 82958 and SKF 77434. It was found that both of these D₁ agonists are readily self-administered by rats. Interestingly, this finding contradicts results found by Grech et al. (1996) where SKF 77434 failed to maintain self-administration behaviour in squirrel monkeys. A limitation of the aforementioned research is that, due to the limited selectivity of the pharmacological agents available for the D₁ receptor, it is not possible to confidently attribute the reinforcing effects of these agonist drugs to the actions of the D₁ receptor alone. In order to address this Self et al. (1996) examined whether SKF 82958 (D₁ agonist) is mediated by D₁ or D₂ receptors. Of the D₁ agonists that have been discussed so far SKF 38393, the agent that was not able to maintain self-administration behaviour in rhesus monkeys (Woolverton et al., 1984) actually has the greatest selectivity for the D₁ receptor (compared to D₂ receptors) than either SKF 77434 or SKF 82958, which have

successfully been shown to maintain self-administration behaviour. This indicates that it could potentially be the actions of the D₂ receptor that are mediating the reinforcing properties of these drugs rather than D₁ actions. Self et al. (1996) investigated this by pre-treating rats with either the D₂ receptor selective antagonist raclopride, or the selective D₁ receptor antagonist SCH 23390. A significant increase in antagonist-induced compensatory SKF 82958 self-administration behaviour was found for animals pre-treated with the D₁ receptor antagonist, but not for animals in the raclopride condition. Increases in self-administration responding following administration of an antagonist is believed to reflect a compensatory response. As the antagonist reduces the rewarding properties of the drug, animals then increase their responding in order to try to obtain more reinforcement.

The above-mentioned studies have all been tested on animals with previous drug experience; however, research has also been done to examine self-administration of D₁ agonists in drug naïve animals. It has been found that the D₁-like agonist SKF 82958 will be successfully self-administered by drug naïve animals, whereas D₂ receptor agonists will only be self-administered if the animal has previously been trained on another drug, like cocaine (Self, 2010). This indicates that D₁ receptors may be more important for primary reward than D₂ dopamine receptors.

A large majority of the previously discussed research illustrates that D₁ dopamine agonists have the ability to substitute in for psychostimulants, such as cocaine and methamphetamine, in a self-administration paradigm. This indicates that the reinforcing properties of cocaine and methamphetamine may be due to the fact that they function as indirect dopamine receptor agonists (Self, 2010). Researchers have also been interested in the effects of antagonist pre-treatment on drug self-administration. In line with agonist research, findings indicate that if rats are pre-treated with a D₁-like selective antagonist they will increase their responding for cocaine in a compensatory manner (Self, 2010). This further

confirms the theory that D₁ dopamine receptors are involved in the reinforcing effects of cocaine. For example, Hubner and Moreton (1991) examined the effect of a selective D₁ and D₂ receptor antagonist (SCH 23390 and spiperone, respectively) on cocaine self-administration behaviour in rats. The animals had been trained to lever press on an FR5 schedule in order to receive a 0.1 ml infusion of cocaine (0.75 mg/kg/injection). Results indicated that pre-treatment with both SCH 23390 and spiperone dose dependently increased responding for cocaine, up to a dose of 10mg/kg. This implies that both D₁ and D₂ receptors may be involved in mediating the reinforcing effects of cocaine. More specific findings have illustrated that it is particularly the D₁ receptors in the nucleus accumbens that are important for the reinforcing effects of cocaine. Maldonado, Robledo, Chover, Caine, and Koob (1993) injected SCH 23390 into the nucleus accumbens of rats trained to self-administer cocaine and found that responding was significantly increased. However, when the D₁ antagonist was injected into other specific brain areas such as the posterior caudate nucleus, no significant alteration in responding was found.

These findings are certainly informative, and clearly implicate dopamine in the reinforcing effects of cocaine and methamphetamine. Taken together they also provide a very solid argument for the theory that D₁ dopamine receptors specifically are important in the reinforcing properties of these psychostimulants. However, previous D₁ receptor research is limited by the lack of selective pharmacological agents. Although a number of supposedly selective D₁ agonist and antagonist drugs have been developed, there are currently no pharmacological agents that work solely on the D₁ receptor (Pierce & Kumaresan, 2006). Much research in this field has used the drug SCH 23390, which is arguably the most selective D₁ antagonist available. However, it has been found that SCH 23390 also exhibits a high affinity for D₅ receptors, as well as for serotonin 5-HT₂ receptors (Bischoff, Heinrich, Sonntag, & Krauss, 1986). As mentioned earlier, D₁ and D₅ receptors are distinct. They are

differentially localised throughout the brain and exhibit distinguishable subcellular distribution patterns (Undieh, 2010). Therefore, although they belong to the same family, they are individual receptors and are likely to perform different functions within the brain. Thus, it is important to use an approach that can differentiate between the two. Conclusions about the role of the D₁ receptor derived from research using a pharmacological approach are consequently limited, as the effects may be due to receptors other than D₁.

To address this issue, the current study will be using a genetically altered mutant rat model. Advances have recently been made in gene targeting within laboratory rodent models, which has led to the development of the D₁ mutant rat. Smits et al. (2006), using target-selected N-ethyl-N-nitrosurea (ENU) driven mutagenesis, were able to discover a genetic point mutation where the single amino acid isoleucine was changed in the third transmembrane domain to a serine. This point mutation is believed to critically affect the functioning of dopamine D₁ receptors brain wide. Initial research into this point mutation has indicated that it leads to around a 50% reduction in the membrane expression of D₁ receptors (unpublished data). The major advantage of using this model in D₁ research is that the mutation is selective for the D₁ receptor while leaving the functionality of other receptor subtypes, such as the D₅ receptor, unchanged.

Due to the presence of pluripotent embryonic stem cells in particular mouse strains, genetic mouse models have been available for research longer than genetic rat models, which have been more difficult to produce. Recent research into cocaine using D₁ receptor knock-out mice has found that cocaine does not function as an effective positive reinforcer in these animals (Caine et al., 2007). Compared to wild type mice there was a dramatic decrease in the amount of knock-out mice that met acquisition criteria for cocaine self-administration, with only two out of 23 animals successfully meeting criteria, compared to 84% of the wild type population. This effect was found to generalise across animals with differing

experimental and breeding histories. Caine et al. (2007) also examined the ability of other agents, namely opioid and food, to exert reinforcing effects in these animals and found that responding was reliably maintained under these conditions in D₁ knock-outs and wild type controls. This illustrates that these animals do not have a general impairment regarding positive reinforcement, and allows the authors to reliably conclude that the specific mutation of the dopamine D₁ receptor in these mice eliminates the reinforcing effects of cocaine. In conjunction with this conclusion is research that has also examined cocaine self-administration in D₂ receptor mutant mice. Caine et al. (2002) used mutant mice lacking the D₂ receptor and found that these animals will readily self-administer cocaine in an almost identical fashion to wild type controls. This finding indicates that the D₂ receptor is not important for the reinforcing effects of cocaine and taken in combination with the previous study supports the idea that D₁ and D₂ dopamine receptors have dissociative roles in cocaine reinforcement, with the D₁ receptor being more important (at least with respect to cocaine and in the mouse).

Unfortunately to date it does not appear that studies using genetic mouse (or other animal) models have examined methamphetamine or other amphetamine-like self-administration. The majority of research in this area is done using cocaine as this drug has proven to be very readily self-administered by animals. In addition, cocaine use and abuse is more problematic in the United States than that of amphetamine type drugs. In a recent national health survey cocaine was found to be the third most prevalent type of drug used within the United States with life time prevalence estimated to be at around 14.7%. In contrast, the lifetime prevalence of amphetamine use, was reported at 8.5% (Ciccarone, 2011). For these reasons, cocaine is often the preferred choice in self-administration research. Interestingly, in terms of pervasiveness the reverse is true in New Zealand with the 2008 national drug survey revealing that amphetamines were the fourth most prevalent drug

used for recreational purposes and also the most commonly used stimulant drug, while cocaine is used much less.

While the literature reviewed thus far indicates the involvement of the D₁ dopamine receptor in the reinforcement processes of cocaine and methamphetamine, of particular interest to the current study is whether the role of the D₁ receptor in these processes is dissociable. Both cocaine and methamphetamine are psychostimulants that exert their reinforcing effects via the dopamine system, however, the way in which they do this differs (Giros, Jaber, Jones, Wightman, & Caron, 1996).

While cocaine is known to have inhibitory effects on serotonin, norepinephrine, and dopamine nerve terminal transporters, research indicates that the primary mechanism through which cocaine exerts its' reinforcing effects is dopamine transporter inhibition. Cocaine effectively blocks the dopamine transporter meaning that the neurotransmitter cannot be taken back up into the neuron. This leads to increased dopamine in the synapse and subsequently, increased stimulation of post-synaptic dopamine receptors. For a long time this was simply assumed, without sufficient binding data available to back it up. However, Ritz, Lamb, and Kuhar (1987) provided evidence to support this theory by directly comparing the relationship between cocaine self-administration in rodents, and dopamine transporter binding. Using [3H]mazindol binding it was found that the potency with which cocaine inhibited binding at the dopamine transporter, positively correlated to the effectiveness of cocaine to produce self-administration. A large number of other pre- and post-synaptic binding sites were also tested but no significant correlations were found, supporting the idea that dopamine uptake inhibition is the primary mediator of the reinforcing effects of cocaine.

In comparison, amphetamine-like drugs, such as methamphetamine, work on dopamine in a more complex manner. Like cocaine, methamphetamine works to some extent

on all monoamine neurotransmitters but the reinforcing effects of methamphetamine are also most frequently associated with the dopamine transporter (Fleckenstein, Volz, Riddle, Gibb, & Hanson, 2007). Instead of simply blocking the dopamine transporter as cocaine does, methamphetamine works to reverse transporter action, meaning that rather than taking neurotransmitter back up into the neuron, the transporter releases more dopamine into the synapse. A study by Krasnova et al. (2010) examined the effect that methamphetamine self-administration had on brain monoamine systems. An extended access model of self-administration was used and animals were given access to methamphetamine for 15 hours a day over a period of eight days. Following this, animals were euthanised either 24 hours, 7 days or 14 days after drug access was taken away. It was found that animals progressively increased their drug intake over the eight-day period and this drug taking behaviour was found to result in a dose-dependent decrease in cortical and striatal dopamine levels. Levels of cortical and striatal dopamine transporter proteins were also found to be depleted. Methamphetamine use was also found to affect serotonin and norepinephrine systems, though this was only in a temporary manner; small depletions were found in animals' euthanised at 24 hours, however, these levels had returned to normal in the brains of rats' euthanised 7 days after drug cessation. These findings clearly indicate that the dopamine system, and in particular the dopamine transporter, are crucially involved in methamphetamine use.

In addition to having differences in mechanisms of action, studies have also shown that cocaine and methamphetamine act on different sites of the dopamine transporter. This idea was suggested by Ritz et al. (1987) in their previously discussed cocaine binding study. Due to the fact that amphetamines generalise to cocaine in animal drug discrimination studies (Huang & Wilson, 1986), it was thought that for amphetamine reinforcement, binding to the dopamine transporter was likewise important. However, findings indicated that both the amount of [^3H]mazindol binding inhibition and self-administration potency that occurred

after amphetamine administration differed from cocaine. The effectiveness of *d*-amphetamine self-administration was found to be much greater than would be predicted based on the weak potency of dopamine transporter binding found. From this it was suggested that the molecular mechanisms of action of cocaine and amphetamine are dissociable. Wayment, Meiergerd, and Schenk (1998) investigated this further by examining the differences in amphetamine and cocaine binding sites on the dopamine transporter using a kinetic model. Results indicated that amphetamine and cocaine do bind to the transporter at different sites. Amphetamine was shown to bind to a common site (the same site that dopamine interacts with) while cocaine acted at a separate interactive molecular site.

Crystallographic methods have also been used to identify the exact binding pockets for these drugs within the dopamine transporter (Cheng et al., 2015; Wang, Penmatsa, & Gouaux, 2015). Overall these studies support the previous mentioned finding that cocaine and amphetamine bind to different sites on the transporter. These studies also add that these drugs lead to different changes in the three-dimensional structure of the transporter, likely the reason underlying their differential mechanisms of action - that is: blockade, as compared to reversal.

This research suggests that the binding profiles of cocaine and amphetamines are different. Selective genetic mutation studies have also examined which specific residues in the dopamine transporter are involved in cocaine and amphetamine binding. Wu and Gu (2003) recently discovered that residue phenylalanine 105 located in transmembrane domain two of the dopamine transporter in mice was highly important for cocaine binding. When residue 105 was switched to cysteine a mutant mouse model was created with decreased cocaine sensitivity but with retained transporter functionality. This is an exciting finding as it suggests that cocaine affinity can be manipulated without affecting dopamine uptake. To extend this research and also investigate the effects of selective genetic mutations on

amphetamine drugs Chen, Han, and Gu (2005) explored whether mutations effecting other residues that surround phenylalanine 105 would also influence the effectiveness of cocaine. Three rounds of sequential random mutagenesis were performed and a successful triple mutation was discovered. This mutation was shown to significantly decrease cocaine sensitivity to a higher degree than the previous phenylalanine 105 mutation. Mutant mice with this triple mutation were found to have a 69-fold decrease in cocaine sensitivity compared to wild type animals. Interestingly, this mutation did not significantly affect the binding of amphetamine to the dopamine transporter. This provides more support for the theory that cocaine and amphetamine bind to and exert their effects on different sites of the dopamine transporter.

As well as exerting their effects differently on the dopamine transporter, cocaine and methamphetamine are believed to work on different pools of dopamine. Numerous studies have illustrated that cocaine releases dopamine from alpha-methyl-*para*-tyrosine-resistant, reserpine-sensitive pools, whereas amphetamine (and by analogy methamphetamine) works on alpha-methyl-*para*-tyrosine-sensitive, reserpine-resistant pools (Verheij & Cools, 2008). Based on these findings, and the clear dissociation in the way methamphetamine and cocaine influence dopamine neurotransmission, it is important to investigate the role of the D₁ receptor in cocaine and methamphetamine.

While self-administration has long been regarded as one of the most successful and best analogues of human drug taking in animal research, a limitation the current study faces is the fact that D₁ mutant Wistar rats do not appear to have the ability to lever press or perform other appropriate operant responses. Previous work in our lab has shown that D1 mutant rats fail to learn a simple autoshaping experiment where they were required to poke their nose through a hole in order to get a delivery of sweetened condensed milk. A nose-poke response was examined as it is a natural exploratory behaviour that should be easily performed by all

rodents. However, compared to the wild type controls who were able to learn the operant nose-poke response, D₁ animals exhibited a major deficit in reinforcement learning. Similar studies with very similar results were done using lever presses as operant responding for reward. There are a number of possible explanations for why these animals will not perform simple operant behaviours in order to receive reinforcement. They may have slight motor impairments due to their depleted dopaminergic system or they may not be motivated enough to work for reward. While this issue is interesting in itself, what this means for the current study is that we are unable to examine the reinforcing effects of methamphetamine and cocaine in D₁ mutant animals using a self-administration paradigm. Instead, the current study uses a battery of other experimental assays to examine different aspects of methamphetamine and cocaine and their reinforcing properties. These tests include a detailed analysis of methamphetamine- and cocaine-induced locomotor activity, conditioned taste avoidance, and finally conditioned place preference. For each experiment both D₁ mutant and wild type control animals are tested under either methamphetamine, cocaine or saline conditions using a range of doses.

Firstly, in order to ascertain the behavioural effects of cocaine and methamphetamine in D₁ mutant rodents, as compared to wild type control animals, we ran a locomotor activity experiment. A theory within the drug addiction field states that a common factor underlying drugs of abuse is their ability to induce motor activation (Zubrycki, Giordano, & Sanberg, 1990). A large number of studies have clearly shown that cocaine and methamphetamine have the ability to induce locomotor activity in rats. Psychostimulants induce locomotor activity through stimulation of the dopamine system; dopaminergic neurotransmission is enhanced due to inhibition and/or reversal of the dopamine transporter which leads to increased concentration of synaptic dopamine (Kita, Shiratani, Takenouchi, Fukuzako, & Takigawa, 1999). Researchers have examined the effects of dopamine antagonist agents on

locomotor activity in rats. Kita et al. (1999) pre-treated Wistar rats with either SCH 23390 or nemonapride (a selective D₁ and D₂ receptor antagonist, respectively) then measured cocaine-induced locomotor activity. Findings illustrated that cocaine reliably increased locomotor activity in control animals, however, SCH 23390 and nemonapride pre-treatment lead to a significant reduction in activity. More specifically Hall, Powers, and Gulley (2009) illustrated that localised administration of SCH 23390 into the medial pre-frontal cortex produced significant dose-dependent decreases in both amphetamine and methamphetamine-induced locomotor activity. Similar research has also been carried out using D₁ receptor mutant mice. Both amphetamine and cocaine were studied and findings illustrated that after both acute and repeated cocaine administration, locomotor responses were significantly diminished in mutants compared to wild type controls. Acute administration of amphetamine was also found to lead to reduced amphetamine-induced locomotor activity in D₁ mice compared to wild types. In contrast to this, the D₁ mice appeared to behave similarly to wild type mice after repeated exposure to amphetamine. However, the authors also discovered that repeated saline injections likewise lead to an increase in locomotor activity in D₁ mutants and therefore this result could possibly be explained in part due to stress-induced sensitisation as a result of injection (Xu, Guo, Vorhees, & Zhang, 2000).

While the aforementioned research indicates a role for the D₁ receptor in methamphetamine- and cocaine-induced locomotor activity in general, the current study aims to provide a more in depth analysis of the behavioural effects of these drugs. The majority of research in this area has focused solely on the forward locomotion of an animal using infrared beams to assess the position of the animal. Some studies have also analysed some additional behaviours (most notably rearing), but the infrared beam methodology does not allow for a more detailed analysis of motor patterns (Fray, Sahakian, Robbins, Koob, & Iversen, 1980). Locomotor behaviour is influenced by a number of different processes and in itself is multi-

dimensional (Paulus, Dulawa, Ralph, & Geyer, 1999). In order to characterise how many independent factors are involved in the unconditioned motor activity of rats, Paulus and Geyer (1993) performed a detailed measurement of motor activity and then carried out a factor analysis. Findings indicated three independent factors, the first was general amount of activity, the second referred to the organisation of sequential responses, and the last factor was exploratory activity. This finding supports the idea that simple infrared beam methodology is not sufficient for analysing all aspects of locomotor activity in rats and emphasises the need for a more detailed approach.

Following this, detailed analyses were carried out by Antoniou, Kafetzopoulos, Papadopoulou-Daifoti, Hyphantis, and Marselos (1998) who performed a comparative study examining the effect of *d*-amphetamine, cocaine and caffeine on a large repertoire of behavioural patterns in rats. Behaviours analysed included, but were not limited to, sniffing, rearing, grooming, scratching, freezing, head swinging and licking. Results for amphetamine indicated that at low to moderate doses the drug generally increased the occurrence of many behavioural responses and also lead to a large amount of switching between behaviours. However, at high doses animals no longer performed frequent behavioural switching and instead, a strong pattern of stereotypy or response repetition emerged. For cocaine results were similar in some aspects to amphetamine, however, the behavioural profiles differed in the fact that cocaine administration mainly leads to increased motor activation rather than the stereotypy produced by amphetamine. The current study aimed to extend this work by examining the specific behavioural profiles of cocaine and methamphetamine in D₁ mutant rodents. By examining these behaviours in mutant animals, we are able to obtain some insight into the role of the D₁ receptor in these behaviours when under the influence of cocaine or methamphetamine.

The second experiment we performed was conditioned taste avoidance (CTA). Avoidance of a rewarding solution such as saccharin or sucrose conditioned through drugs of abuse is thought to illustrate dopamine-dependent and appetitive properties of a drug (Fenu, Cadoni, & Di Chiara, 2010). This paradigm was originally known as conditioned taste aversion and illustrated that rats would avoid a rewarding sucrose or saccharin solution that had been paired with a substance causing gastrointestinal distress, such as lithium chloride (Parker, 1995). However, further research into this assay discovered the paradoxical effect that drugs of abuse, readily self-administered by animals, will also lead to a taste aversion. For example Wise, Yokel, and DeWit (1976) trained rats to lever press for infusions of apomorphine, in conjunction with pairing a saccharin solution with apomorphine. Results illustrated that animals would avoid the drug-paired saccharin solution, but still continued to lever press in order to self-administer apomorphine. These findings provided some of the first evidence that drugs of abuse potentially have the ability to produce both positive and negative effects simultaneously. Subsequent research has gone on to show that a large range of drugs of abuse from different drug classes, such as morphine, cocaine, nicotine and amphetamine, will produce CTA (Fenu et al., 2010). Original theories put forward to explain this phenomenon suggested that drug-induced CTA occurs as a result of the aversive properties of the drug (especially after the initial reinforcing effects wear off). Due to the similarities between drug-induced and classical toxin-induced CTA, the negative effects produced by drugs of abuse were posited to be similar in nature to those of classical toxins, such as lithium chloride (Verendeev & Riley, 2012). However, work investigating this idea failed to find supporting evidence. Grill and Norgren (1978) developed the taste reactivity test which is an assay used to assess responses to gustatory stimuli. It was discovered that rats exhibit very distinctive orofacial expressions or reactions (such as chin rubbing, gaping or face washing) after tasting strongly flavoured solutions. How an animal reacts to a certain solution can be

identified as either appetitive or aversive based on the ingestive or rejective nature of their reaction, respectively (Mucha, 2010). Using the taste reactivity test Parker (1995) assessed the difference between CTA produced by drugs of abuse (including but not limited to, amphetamine, cocaine, methamphetamine, morphine and nicotine) compared to the emetic drug lithium. Results showed that there was a qualitative difference in the aversive responses caused by these drugs. While both lithium and the drugs of abuse successfully produced CTA at similar doses, findings from the taste reactivity test found that animals exhibited strong aversive responses following lithium administration, but these were not present following administration of rewarding drugs. Based on these findings, researchers began to refer to rewarding drug-induced CTA as conditioned taste avoidance as compared to emetic-induced condition taste aversion.

While work has been done examining the role of the dopamine D₁ receptor in CTA learning, this has for the most part only been done in lithium chloride-induced CTA. Findings from Fenu, Bassareo, and Di Chiara (2001) illustrated that D₁ antagonists SCH 23390 and SCH 39166, when given in a time appropriate manner, diminished the acquisition of CTA. However, administration of the D₂/D₃ antagonist raclopride did not significantly impair CTA learning. These findings indicate a critical role of the D₁ receptor in lithium chloride-induced CTA learning, although, as it is believed that drug and lithium-induced CTA occur as a result of different underlying processes, these results should be applied lightly to the current study.

A small number of studies have examined the role of dopamine receptors in drug-induced CTA. Hunt, Switzman, and Amit (1985) found that pimozide would decrease cocaine-induced saccharin avoidance. While this indicates that dopamine plays a role in cocaine CTA, the use of pimozide does not allow for any indication of the precise role of the D₁ receptor in this paradigm. More specific research using D₁ receptor antagonists does not appear to have been carried out using cocaine. However, researching assessing the effects of

both SCH 23390 and raclopride on *d*-amphetamine CTA has been carried out and illustrated that when both antagonists were administered in conjunction, saccharin avoidance was diminished. However, if SCH 23390 was given independently, saccharin intake was not significantly affected, indicating that the D₁ receptor alone may not mediate the processes underlying CTA (Lin, McGregor, Atrens, Christie, & Jackson, 1994). One particular study carried out by Fenu, Rivas, and Di Chiara (2005) directly compared the role of dopamine in lithium and drug-induced CTA, and found that while SCH 39166 administration impaired both nicotine and morphine CTA, SCH 39166 given under the same conditions did not affect lithium-induced saccharin avoidance. These findings indicate that the role of dopamine differs between CTA-induced by drugs of abuse compared to lithium. These findings provide support for the theory that the processes underlying these two different forms of CTA are dissociable, and in turn provides support for the idea that drug-induced CTA occurs due to the rewarding properties of the drug. This therefore makes CTA an appropriate paradigm with which to measure appetitive properties of drugs (Fenu et al., 2005). These findings also highlight a potential difference between nicotine/morphine-induced CTA and cocaine-induced CTA, as a D₁ antagonist alone is shown to block taste avoidance conditioned through nicotine or morphine pairings, but both D₁ and D₂ receptor functioning needs to be impaired for cocaine-induced CTA to be reversed.

The final experiment that we ran in the current study was conditioned place preference (CPP). CPP is a classic assay for measuring drug reinforcement and has been used in a magnitude of studies to illustrate that when a distinctive environment is paired with drugs of abuse animals will develop a preference for that environment, compared to a dissociable environment which has been paired with saline (Prus, James, & Rosecrans, 2009).

Due to the fact that CPP measures the rewarding properties of a drug and most drugs of abuse are thought to exert their rewarding effects through dopamine, work with dopamine

agonists has previously been carried out using the CPP assay. Based on earlier work illustrating that D₁-like and D₂-like agonists appear to have different effects on self-administration, Graham, Hoppenot, Hendryx, and Self (2007) examined the effect of both D₁ and D₂ agonists to induce CPP, often argued to be a similar paradigm. It was found that in drug naïve rats' three different D₁ receptor agonists (ABT-431, SKF 81297, and SKF 82958) dose-dependently produced CPP. However, D₂ agonists had no effect. This finding provides more support for the idea that D₁ but not D₂ receptors are important for mediating primary reward, and also implies that they are involved in the CPP process.

The effect of D₁ receptor blockade on drug-induced CPP acquisition has also been investigated. Both SCH 23390 and SCH 39166 have been shown, at low doses, to completely eliminate place preference conditioned through amphetamine (Acquas & Di Chiara, 1994; Hiroi & White, 1991). A more specific role for the D₁ receptor in amphetamine-induced CPP was examined by Hiroi and White (1991) who looked at how both D₁ and D₂ receptors were involved in amphetamine CPP, both generally throughout the brain, and specifically in the nucleus accumbens and striatum. It was found that when the antagonists were administered prior to conditioning, both D₁ and D₂ agents successfully blocked CPP acquisition. However, when these drugs were administered systemically after conditioning had already taken place and before testing, only the D₁ receptor antagonist successfully, and dose-dependently, diminished the expression of previously conditioned place preference behaviour. Using microinjections, it was also found that only injection of antagonists into the nucleus accumbens, but not the striatum, eliminated CPP expression. These results suggest that both D₁ and D₂ receptors are important for establishing amphetamine-induced CPP. However, expression of place preference after conditioning appears to be mediated primarily through activation of dopamine D₁ receptors, specifically in the nucleus accumbens.

Work examining the role of D₁ receptors in cocaine-induced CPP is also prevalent within the literature, and illustrates that D₁ antagonist administration blocks cocaine paired place preference (Baker, Fuchs, Specio, Khroyan, & Neisewander, 1998; Cervo & Samanin, 1995; Arbi Nazarian, Scott J. Russo, Eugene D. Festa, Mohammed Kraish, & Vanya Quinones-Jenab, 2004). Interestingly, research has also been done examining cocaine CPP in apomorphine susceptible and unsusceptible rats. These rats were developed as an animal model for psychopathology research and have shown promise especially as a tool for examining schizophrenia (Ellenbroek & Cools, 2002). As mentioned earlier in this report apomorphine is a non-selective dopamine agonist which stimulates both D₁-like and D₂-like receptors. Therefore, it follows that in animals with apomorphine susceptibility there is a higher amount of dopamine activation occurring, compared to unsusceptible animals. van der Kam, Coolen, Ellenbroek, and Cools (2006) found that apomorphine susceptible rats would develop cocaine-induced CPP at lower doses than unsusceptible animals, indicating that apomorphine unsusceptible animals found cocaine to have a smaller rewarding value. These results imply that activation of both D₁ and D₂ receptors is important for the rewarding properties of cocaine as assayed by CPP.

Aims and Hypotheses

The aim of the current study was to further investigate the role of the dopamine D₁ receptor in reward using a novel D₁ mutant rat model. Specifically, we were interested in how D₁ receptors are involved in the behavioural and reinforcing properties of methamphetamine and cocaine.

Firstly, a locomotor activity experiment was carried out in order to examine the behavioural effects of each of these drugs. It was hypothesised that locomotor responses will

be reduced in D₁ animals compared to wild types across drug doses of both methamphetamine and cocaine.

For the conditioned taste aversion experiment, based on previous antagonist research that indicates both D₁ and D₂ receptors are important for the expression of CTA, it is hypothesised that, due to the fact D₁ mutant rats only have a 50% reduction in D₁ receptor functioning, there will be no difference in taste avoidance found between genotypes for methamphetamine. In terms of cocaine-induced CTA, due to the fact that the literature illustrates differences in the taste aversion of substances conditioned through different drugs of abuse (such as nicotine and heroin) and due to the differences between methamphetamine and cocaine we are currently not sure what results to expect from this condition.

For the conditioned place preference paradigm, it was hypothesised based on work using D₁ antagonist pharmacological agents that, due to the fact D₁ receptors appear to be important for the expression of both cocaine- and amphetamine- induced CPP, D₁ mutant animals will show decreased CPP compared to wild type controls.

In terms of the differences between cocaine and methamphetamine in the above paradigms, it was hypothesised that the D₁ mutants will show more deficits in response to methamphetamine than cocaine, as compared to wild type animals. We hypothesise this based on the different way methamphetamine works on the dopamine transporter compared to cocaine. Because methamphetamine leads to a reversal rather than a blockage of the transporter, this means that dopamine release, and subsequently receptor stimulation, is increased more after methamphetamine than after cocaine treatment. Due to this increased receptor stimulation, effects are likely to be more exaggerated in D₁ mutant rats due to their reduced D₁ receptor expression.

Materials and Method

Animals

For the following experiments female dopamine D₁ mutant Wistar rats and wild type (wild type) Wistar rats were used. All animals were adult (over 70 days) by the time they began experimental testing. The D₁ mutant rats were originally developed using an ENU target-selected mutagenesis technique and the current animals were backcrossed onto wild type rats for at least 10 generations. Animals were bred under controlled conditions and following weaning were housed in standard hanging polycarbonate cages in groups of two to four animals per cage. These cages were located inside the housing rooms located within the vivarium at Victoria University of Wellington. Both D₁ and wild type rats were both housed under the same conditions with the housing rooms being temperature (19-21°C) and humidity (55%) controlled. Rooms were kept on a 12-hour light-dark cycle (lights on 0700 h). All animals had access to food and water *ad libitum* except for when they were subject to any experimental restrictions. All experimental procedures were approved by the Victoria University of Wellington Animal Ethics Committee.

Drugs

Cocaine sulphate (BDG Synthesis, Wellington, New Zealand) and methamphetamine hydrochloride (ESR, Auckland, New Zealand) were used for all experiments. Both cocaine and methamphetamine were dissolved into a sterile solution of 0.9% saline and delivered through intraperitoneal injection. For methamphetamine doses of either 0.25, 0.5 or 1 mg/kg were administered. For cocaine doses were either 2.5, 5, or 10 mg/kg.

Cocaine- and Methamphetamine-Induced Locomotor Activity

Apparatus

During this experiment, animals were removed from their home cages and placed into a round open field apparatus measuring 80cm in diameter. The open field is made of Plexiglas and coloured black in order to create maximum contrast between the background and the white rodents. Animals were recorded using a video camera located above the open field, the video camera fed straight through to the computer and was connected to EthoVision XT v9.0 tracking software that was subsequently used to analyse the videos. EthoVision tracks rodent movement by identifying three body points: nose, middle body and tail point.

Behavioural Measures

In order to examine the detailed behavioural profiles induced by methamphetamine and cocaine five different behaviours were analysed: overall distance travelled, rearing, immobility, rotation, and head weaving. All measures were analysed using EthoVision software.

Overall distance travelled was used as a measure of general locomotor activity and was measured by tracking how far the centre point of the animal moved, in meters, over the duration of the testing session.

Rearing was analysed first by creating a rearing zone. The rearing zone was defined as the outside edge of the open field. An animal was defined to be rearing if the nose point entered into this zone. Both duration of time spent in the rearing zone and frequency of entries were recorded and analysed. Thus only supported rearing was analysed while rearing free from the wall was not.

For immobility both duration and frequency was measured. Immobility was defined as any period when the body points of the rodent were not moving. Duration refers to the total amount of time the animal spent motionless over the entire testing period, whereas frequency measured the amount of times the animal stopped moving.

Frequency of both clockwise and counter clockwise rotation was analysed. Rotation was measured using a turn angle parameter. When one turn angle of the body points by 360 degrees occurred, this equalled one rotation.

Finally, head weaving was measured by analysing the nose turning angle of the animals. Nose turning angle was calculated when the nose point of the animal moved in any direction away from the body point which remained stable.

Procedure

In total 12 wild type and 12 D₁ animals were used in this experiment. All animals were drug naïve prior to the beginning of testing. Due to delays in breeding, this experiment was run in two groups. First, eight wild type and eight D₁ rats were tested these animals were randomly assigned into either a methamphetamine or a cocaine condition. If in the cocaine condition animals would receive, on four separate testing sessions, saline, 2.5 mg/kg, 5 mg/kg, and 10 mg/kg of cocaine given intraperitoneally. If assigned to the methamphetamine condition animals would receive saline, 0.25 mg/kg, 0.5 mg/kg, and 1 mg/kg of methamphetamine given intraperitoneally. The order in which each rat received these drug doses was randomised to control for any order effects that may have occurred. The second group of animals (4 wild type and 4 D₁) run in this experiment were subject to the same conditions, however, they did not receive a saline dose. A wash out period of at least 8 days was implemented in between testing sessions in order to minimise any effects that may have occurred due to repeated administration of drug.

For each testing session, animals were injected (i.p.) with the appropriate dose of drug and then placed into the open field. The behaviour of the rodent was recorded for 30 minutes. These videos were then later analysed in detail using EthoVision software for a number of

specific behaviour patterns including overall locomotor activity (distance travelled), rearing, rotation, immobility and head weaving.

Conditioned Taste Avoidance

Apparatus

Two days prior to the beginning of the experiment animals were transferred out of their group cages and moved separately into individual standard hanging polycarbonate cages. Housing conditions remained the same as before with food available *ad libitum* throughout the experiment, however water access was restricted. White plastic drinking bottles with non-leak stoppers in the drinking tubes were used in order to make sure that fluid intake calculations were as accurate as possible.

Procedure

This experiment was based on the protocol described by Fenu et al. (2010), some aspects have been adapted. Five separate experimental groups of animals were used for this experiment. A control saline condition was run first using eight wild type and eight D₁ animals, this was followed by four separate drug conditions: 0.25mg/kg and 0.5mg/kg of methamphetamine as well as 2.5mg/kg and 5mg/kg of cocaine. Six wild type and six D₁ mutant Wistar rats were used for each of these drug conditions. Overall, a total of 64 animals were used for this set of experiments, prior to testing all rats were drug and experiment naïve. For this reason, the animals were handled on days prior to the testing sessions in order to make injections less stressful for them.

24 hours before the first experimental session animals were water deprived. For each drug group experiments were carried out over a six-day period and consisted of three phases. The first phase was the training phase, which took place over three days. The main aim of this phase was for the animals to learn that they would only get limited access to liquid each

day. Each rat was given water for a 30-minute period. Fluid intake was measured in order to ensure animals were drinking in this period. This was done by weighing water bottles before and after each session. In the second phase (the conditioning phase) animals received 5% sucrose solution for a 30-minute period, fluid intake was again measured by weighing the bottles. Directly after this they were injected (i.p.) with either saline, 0.25 mg/kg methamphetamine, 0.5 mg/kg methamphetamine, 2.5 mg/kg cocaine, or 5mg/kg cocaine. On these days the drinking period was spaced out between D₁ and wild type animals in order to ensure they received their drug injections as soon as possible after their drinking time ended. This phase lasted two-days and the injection order was counter-balanced over days (e.g. on the first day wild type would drink and receive their injections first in order from 1-6, on day two wild type animals would receive their injections second in order from 6-1). On the final day of the experiment (the choice phase) each rat received both 5% sucrose and water for 30 minutes. Fluid intake was measured and the degree of conditioned taste avoidance for each animal was calculated (the percentage of sucrose intake relative to complete fluid intake). During the training and conditioning phases, the placement of the drinking bottle was alternated to ensure that animals did not just learn to drink from one side of the cage. On the final testing day, the order of the sucrose compared to the water solution was alternated between animals.

Methamphetamine- and Cocaine-Induced Conditioned Place Preference

Apparatus

Apparatus for this experiment consisted of eight Plexiglas locomotor activity chambers (Med Associates Inc, USA; model ENV-515 measuring 42 x 42 x 30cm) fixed inside sound and light attenuating boxes. Two-compartment inserts (Med Associates Inc, USA; model ENV-517) were fitted into the chambers in order to create two distinctive

environments that were physically separated from each other. Chambers were equipped with infrared sensors, which were used to track the animal's movement inside the box. Med Associates computer software connected to the boxes was used to analyse the rat's movements and calculate the time animals spent in each area of the box.

Each compartment of the insert was made distinctively different from the other by covering the walls with different patterns (one side lines, one side circles). Differently textured floor inserts (one smooth, one rough) were also used to create contrasting tactile environments. Small lights were fixed above each chamber and these remained on for each testing and conditioning session in order to highlight the differing wall patterns.

Procedure

This experiment was carried out using four different groups of animals (each group contained eight wild type and eight D₁ rats), two groups received methamphetamine and two groups received cocaine – each group was treated with a different drug dose, either 0.5 mg/kg or 1 mg/kg for methamphetamine or 5 mg/kg or 10 mg/kg for cocaine. Each separate group was run over an eight-day period following the same procedure. Unfortunately, due to animal constraints majority of the rats were not drug naive prior to testing, they were used in the previous CTA experiment, however, drug exposure in that experiment was low and a significant amount of time had passed between experiments (at least two full weeks) therefore this is not predicted to be an issue.

Phase one of the experiment, the pre-conditioning phase took part on day one. Animals were first habituated to the experimental room for at least 30 minutes, this occurred throughout all phases of the experiment. A low level of white noise was also played in the room to help calm the animals. After this, the rats were placed into the chamber with free access to each compartment for a 15-minute period. The side that the animal was put into the

chamber was counterbalanced across subjects in order to control for side bias that occurred due to where they entered. During the 15-minute period, while the rat was in the box Med Associates computer software tracked the animal's movements. Once the animals had finished they were returned back to the housing room. No drug was administered during the pre-conditioning phase. Med Associates data was subsequently analysed in order to determine how much time each animal spent in each compartment. Once this was calculated the animals least preferred compartment (compartment they spent the least amount of time in) was conditioned through drug pairings. Unfortunately, an un-biased approach could not be used for this experiment as the protocol that has been developed in our laboratory for un-biased pre-conditioning results in only around 60% of animals exhibiting an unbiased preference. Given the constraints of the number of animals this was not a viable option for the current study.

The second phase of this experiment was the conditioning phase, which took place over six days. During this phase animals would be injected intraperitoneally with the appropriate dose of either methamphetamine or cocaine (depending on which group they had been assigned to) then placed into their originally least-preferred compartment, or they would be injected with saline (i.p.) and placed into their originally preferred compartment (see Figure 1). Animals remained in the chamber, with access between compartments blocked off for 40 minutes. After each session and before the next group of animals was run the chamber floors and walls were thoroughly cleaned with 70% ethanol in order to minimise any side biases that may occur due to scents remaining in the chamber from previous animals.



Figure 1. Conditioning phase layout. This figure illustrates how the conditioning phase of this experiment was set out and what solution animals were administered on each day throughout the phase.

The final phase of this experiment was the post-conditioning phase (test day), which lasted one day and took place on day eight. The post-conditioning phase was run identically to the pre-conditioning phase, no drug was administered and the animals were placed in the chamber for 15 minutes with free access to either side. Movement was tracked and analysed using Med Associated software. Data from the testing session was then analysed for time spent in each chamber. The time animals spent in the drug-paired chamber was compared to their original time spent there in the pre-conditioning phase. An animal was considered to have developed place preference if the time spent in the drug-paired chamber was significantly longer during the testing session as compared to the pre-conditioning day.

Results

Experiment One: Locomotor Activity

Videos that were collected during experimental sessions were compiled into one EthoVision file. Tracking parameters (arena and detection setting) were reviewed and altered for each trial video to obtain optimal offline tracking. Once this was completed, the videos were analysed using EthoVision XT v9.0, using the multiple body-point module for a number of behavioural measures. All behavioural measures were automatically scored using EthoVision software. The following behaviours will be described in the results section: overall distance travelled (locomotor activity), rearing, immobility, rotation and head weaving. Data from different drug groups was analysed separately, a multivariate analysis of variance (MANOVA) was used to examine the differences in behavioural responses across drug dose and genotype conditions.

Methamphetamine

Distance Travelled

A significant main effect of genotype was found for distance travelled ($F(1,44)=12.51, p<.001$), as well as a main effect of dose ($F(3, 44)=13.75, p<.001$). However, there was no significant genotype * dose interaction. Inspection of Figure 2 shows that methamphetamine led to a dose dependent increase in distance travelled, and that this effect was smaller in the D1 mutants. Post hoc Tukey test confirmed this, showing significant differences between 1 mg/kg and all the other doses. Subsequent independent samples t-tests were carried out in order to test the hypothesis that D₁ animals would be less sensitive to the effects of methamphetamine than wild type controls. Distance travelled by D₁ and WT animals was directly compared at both saline and 1 mg/kg methamphetamine doses. While no significant difference was found between genotypes under saline conditions ($t(14)=.21, p=.836$), there was a significant difference between groups when a high dose of methamphetamine was administered, $t(10)=2.23, p=.050$.

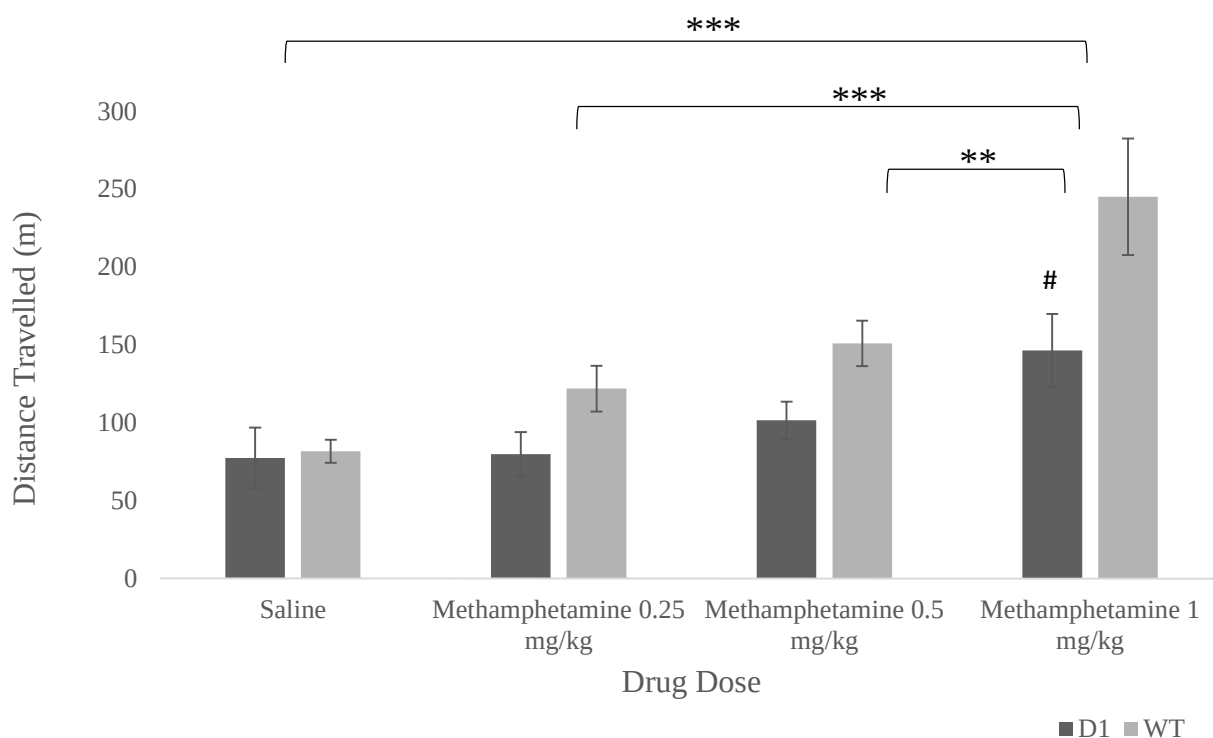


Figure 2. Distance travelled across methamphetamine doses. This graph illustrates the different distances travelled by D₁ and wild type animals across increasing doses of methamphetamine. It is clearly shown that wild type animals travel larger distances than D₁ mutants, and distance travelled in sessions also increases dose-dependently. Error bars represent SEM, ** represents significance at $p<.01$, *** indicates $p<.00$, # indicates a significant genotype difference ($p<.05$).

Rearing

No significant main effect of genotype or dose was found for either duration of time animals spent in the rearing zone of the open field or frequency of entries into the rearing zone. No significant genotype * dose interaction was found.

Immobility - Duration of Immobility

A significant main effect of genotype was found for duration of immobility, $F(1,44)=6.57$, $p=.014$, as well as a significant main effect of dose, $F(3,44)=3.85$, $p=.016$. No genotype * dose interaction was found. As shown in Figure 3, this result indicates that, under methamphetamine conditions, D₁ animals were exhibiting overall a longer duration of immobility during testing sessions than wild type rats regardless of dosage. The main effect of dose was analysed using Post hoc Tukey tests. It was found that for duration of immobility rats exhibited significantly greater periods of motionlessness under saline control than 0.5 mg/kg methamphetamine conditions, $p=.039$. A significant difference in duration of immobility was also found between 0.25 mg/kg and 0.5 mg/kg dose groups, $p=.015$. Subsequent Independent t-test showed that the significant genotype effect was due to a significant difference between wild type and D₁ mutant rats at the 0.25 mg/kg dose ($t(5.62)=2.62$, $p=.042$). Levene's test for equality of variances was significant so equal variances not assumed statistics are reported here.

Immobility – Frequency of Immobility

A significant main effect of genotype was found for frequency of immobility, $F(1,44)=13.53$, $p=.001$, as well as a main effect of dose, $F(3,44)=7.17$, $p=.001$. However, there was no significant genotype * dose interaction. This significant main effect of genotype indicates that wild type animals exhibited much greater stop/start behaviour than D₁ rats (please refer to Figure 4). The nature of the dose effect finding was examined using post hoc

Tukey tests. Analysis revealed a significantly greater number of motionless periods occurring under 0.5 mg/kg, compared to saline ($p = .003$) and 0.25 mg/kg methamphetamine conditions ($p = .013$). Significant differences were also reported between the 1 mg/kg condition and saline ($p = .012$) as well as 0.25 mg/kg groups ($p = .045$). Subsequent independent samples t-tests were carried out in order to test the hypothesis that D₁ animals would be less sensitive to the effects of methamphetamine than wild type controls. Frequency of immobility was compared between D₁ and WT animals under saline and 1 mg/kg conditions. No significant difference was found between genotypes after saline administration ($t(14) = .87, p = .398$) but there was a difference after 1 mg/kg methamphetamine administration, $t(10) = 2.85, p = .017$.

Taken together findings for both duration and frequency of immobility illustrated that while wild type animals exhibit more frequent periods of motionless under methamphetamine conditions, D₁ animals remain immobile for longer durations.

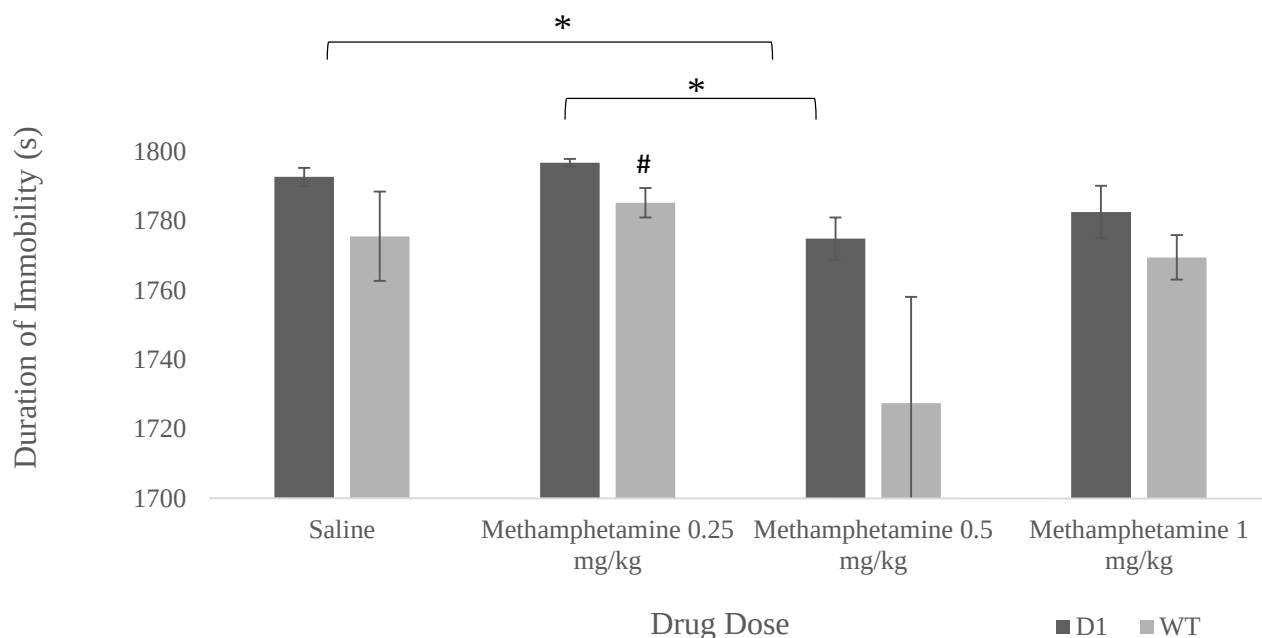


Figure 3. Duration of immobility under saline and methamphetamine conditions. This graph illustrates that under all methamphetamine doses D₁ animals are exhibiting longer periods of motionless. Error bars represent SEM. * Represents significance at $p < .05$ level. # indicates a significant genotype difference ($p < .05$).

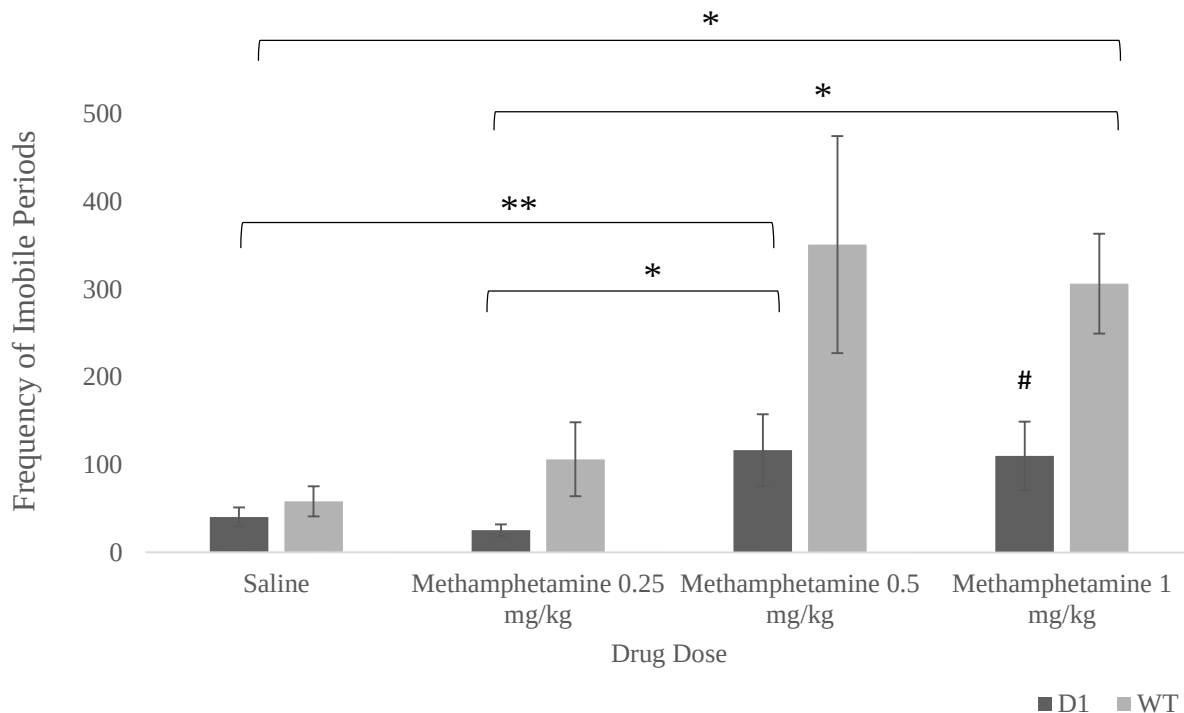


Figure 4. Frequency of immobile periods for D₁ and wild type animals under methamphetamine conditions. This graph illustrates that wild type rats showed a greater frequency of motionless periods than D₁ animals over all methamphetamine doses. Error bars represent SEM, ** represents significance at $p < .01$, * represents significance at $p < .05$ level. # indicates a significant genotype difference ($p < .05$).

Rotation – Clockwise

A significant main effect of genotype was found for clockwise rotation ($F(1,44) = 4.75, p = .035$) as well as a main effect of dose ($F(3,44) = 4.49, p = .008$). No significant genotype * dose interaction was reported. Examination of Figure 5 illustrates that administration of methamphetamine lead to a somewhat dose dependent increase in clockwise rotation, wild type animals also appear to perform a higher number of rotations than D₁ mutants. Post hoc Tukey tests were calculated to confirm this and showed that there was a significant difference between the frequency in clockwise rotation between saline and 1 mg/kg methamphetamine conditions, $p = .010$, as well as between 0.25 mg/kg and 1 mg/kg conditions, $p = .020$. Subsequent Independent t-test showed that the significant genotype effect was due to a significant difference between WT and D₁ mutant rats at the 0.5 mg/kg dose ($t(10) = 2.24, p = .049$).

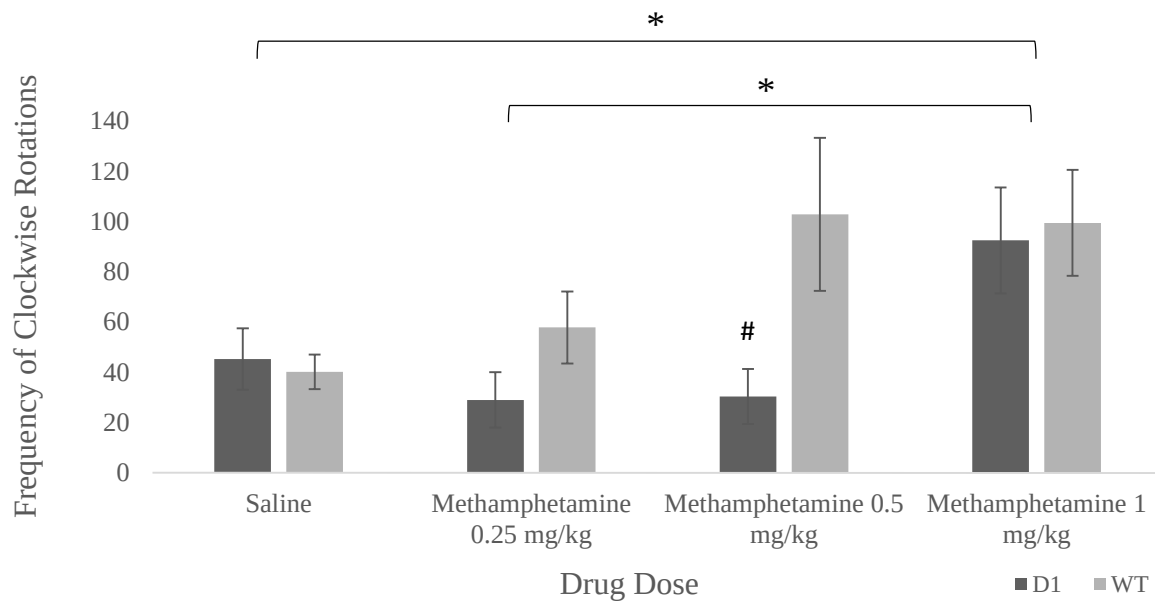


Figure 5. Frequency of clockwise rotations performed by animals under methamphetamine conditions. This data illustrates that at 0.25 and 0.5 mg/kg doses of methamphetamine wild type rats exhibit increased rotation behaviour compared to D₁ mutants. At the highest dose, and under saline conditions, behaviour is comparable across genotypes. Error bars represent SEM, * represents significance at $p < .05$. # indicates a significant genotype difference ($p < .05$).

Rotation – Counter Clockwise

A significant main effect of genotype was found for counter clockwise rotation ($F(1,44) = 4.65$, $p = .037$). A significant main effect of dose was also reported ($F(3,44) = 3.64$, $p = .020$) however, no significant genotype * dose interaction was found. In keeping with what was found for clockwise rotation Figure 6 shows a very similar pattern. The frequency of counter clockwise rotations increases at higher doses of methamphetamine and wild type control rats perform increased rotations compared to D₁ mutants. Post hoc Tukey tests confirmed this showing that under saline conditions the frequency of counter clockwise rotation was significant lower compared to the 1 mg/kg condition, $p = .022$. As shown in Figure 6 the significant genotype effect is likely due to differences between wild type and D₁ mutant rats at the 0.5 mg/kg dose. However, an independent samples t-test assessing this failed to find a significant result, likely due to the variability within the sample.

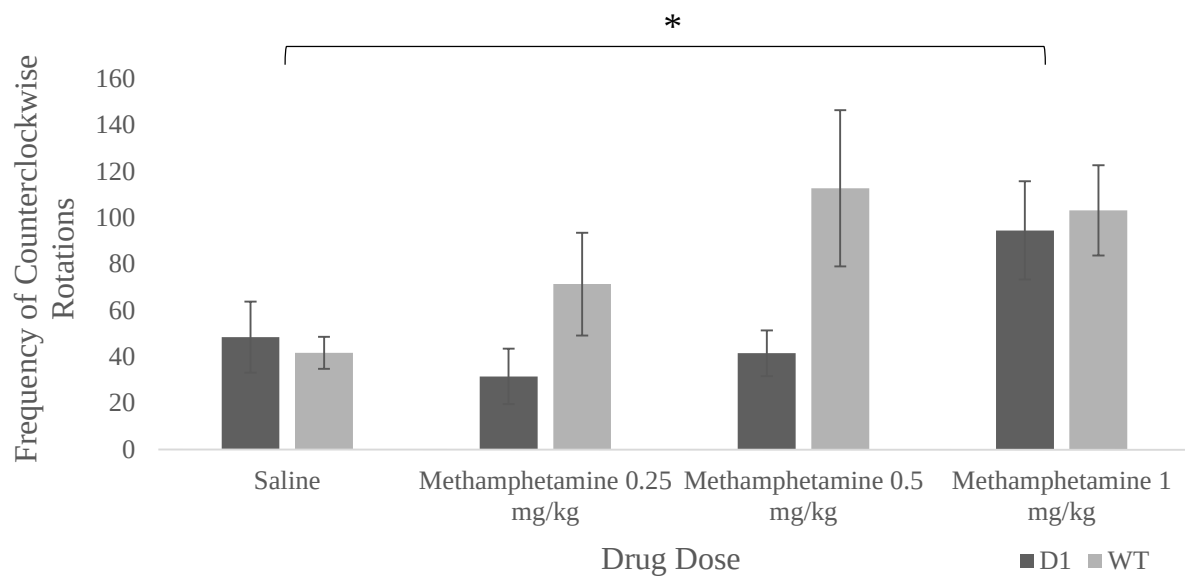


Figure 6. Frequency of counter clockwise rotations performed by animals under methamphetamine conditions. This data illustrates that at 0.25 and 0.5 mg/kg doses of methamphetamine wild type rats exhibit increased rotation behaviour compared to D₁ mutants. At the highest dose, and under saline conditions, behaviour is comparable across genotypes. Error bars represent SEM, * represents significance at $p < .05$.

Head Weaving

A significant main effect of genotype ($F(1, 44) = 5.28, p = .026$) and dose ($F(3, 44) = 11.03, p < .001$) was found for head weaving, but no genotype * dose interaction. This indicates, as shown in Figure 7, that overall dopamine D₁ receptor mutant rodents displayed a higher nose-turning angle than wild types. Post hoc Tukey tests indicate that head weaving under saline conditions was significantly greater than at doses of 0.5 mg/kg ($p = .003$) or 1 mg/kg methamphetamine ($p < .001$). A significant difference was also reported between 1 mg/kg and 0.25 mg/kg dose groups ($p = .016$). Independent t-test showed that the significant genotype effect was due to a significant difference between wild type and D₁ mutant rats at the 0.5 mg/kg dose ($t(10) = 2.44, p = .035$).

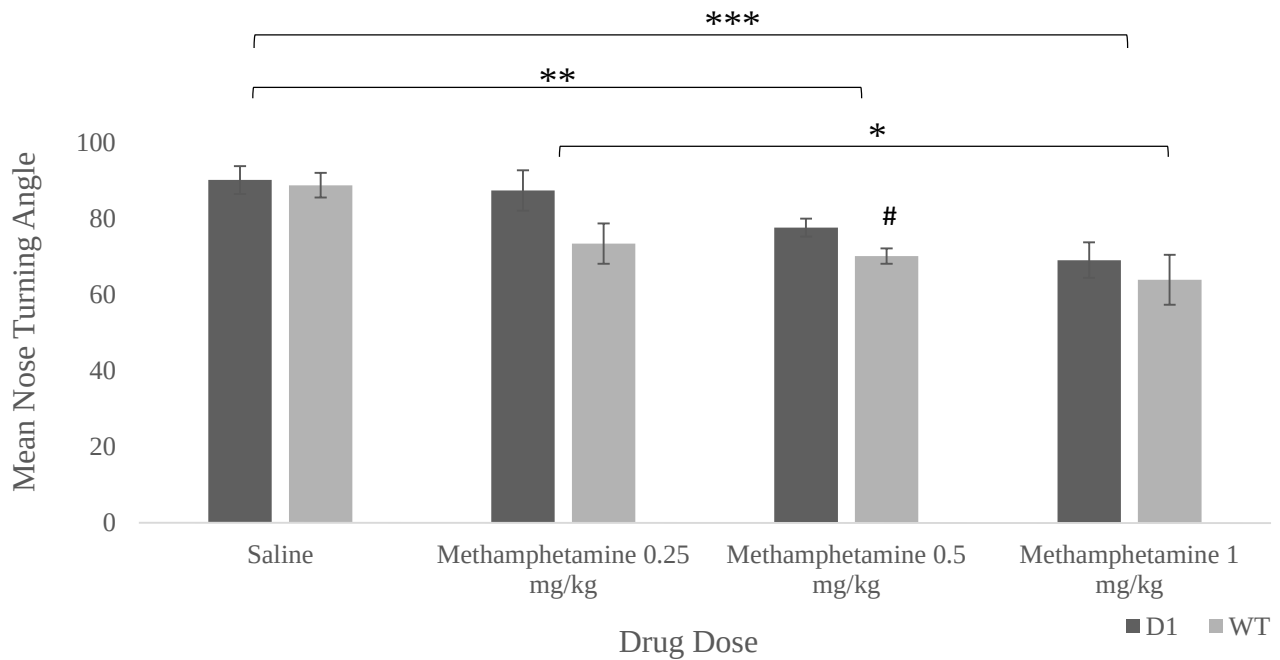


Figure 7. Mean nose-turning angle in animals across methamphetamine drug conditions. This graph illustrates that overall D₁ animals are exhibiting a greater nose turn angle than wild types. Error bars represent SEM. *** represents significance at $p < .001$, ** represents significance at $p < .01$, * represents significance at $p < .05$ level. # indicates a significant genotype difference ($p < .05$).

Cocaine

Distance Travelled

A significant main effect of genotype was found for distance travelled, $F(1, 44) = 7.59$, $p = .008$, indicating that wild type animals exhibit greater levels of locomotor activity than D₁ mutant rodents, as shown in Figure 8. A significant main effect of dose was also found for distance travelled, $F(3, 44) = 6.87$, $p = .001$. Post hoc Tukey tests illustrated that there was a significant difference between distance travelled under saline and 10 mg/kg conditions ($p = .001$) as well as 2.5 mg/kg and 10 mg/kg dose groups ($p = .004$). No significant genotype * dose interaction was found. Subsequent independent samples t-tests were carried out in order to test the hypothesis that D₁ animals would be less sensitive to the effects of cocaine than wild type controls. t-test showed that the significant genotype effect was due to a significant difference between WT and D₁ mutant rats at the 5mg/kg dose ($t(10) = 2.18$, $p = .054$).

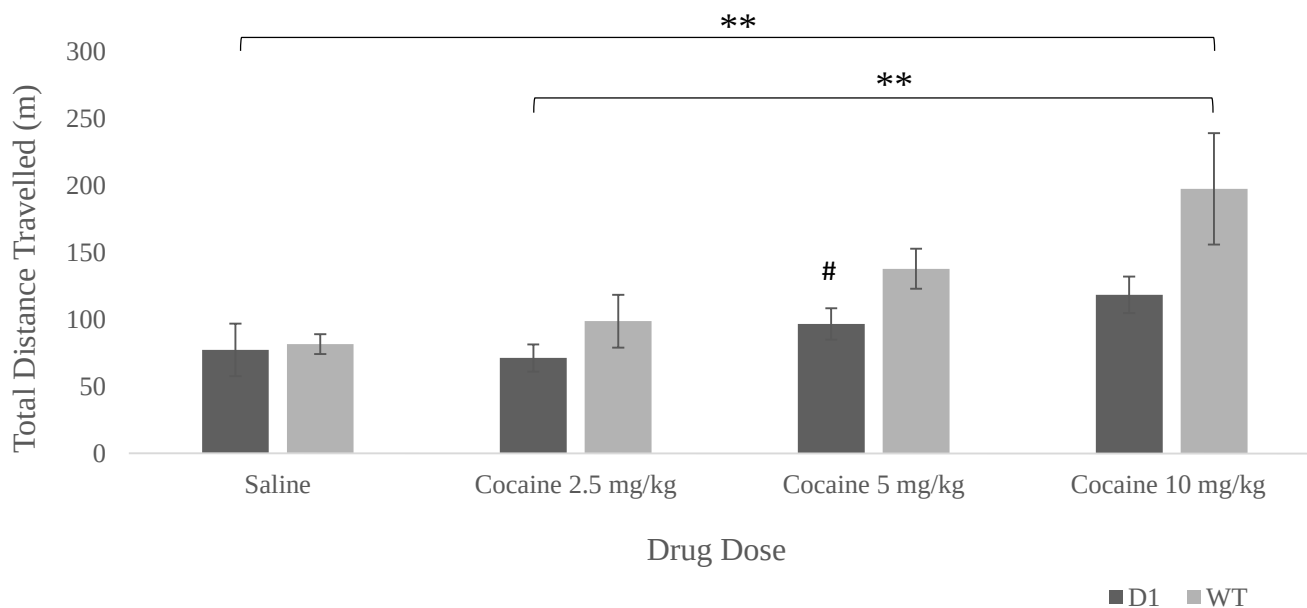


Figure 8. Total distance travelled over cocaine doses. This graph illustrates the total distance travelled in meters by both D₁ and wild type rats over different cocaine dose conditions. Wild type animals appear to travel greater distances across all drug doses with the most distinctive difference between genotypes occurring at the highest 10 mg/kg drug dose. Error bars represent SEM, ** represents significance at $p < .01$ level. # indicates a significant genotype difference ($p < .05$).

Rearing

No significant main effect of genotype or dose was found for either duration of time animals spent in the rearing zone, or frequency of entries into the rearing zone. No significant genotype * dose interaction was found.

Immobility – Duration of Immobility

No significant main effect of genotype or dose was found for duration of time spent motionless under cocaine conditions. No significant genotype * dose interaction was found.

Immobility – Frequency of Immobility

No significant main effect of genotype was found, however, a significant main effect of dose was found for frequency of immobility, $F(3,44) = 4.34$, $p = .009$. Post hoc Tukey tests indicated that the frequency of motionless periods was significantly lower under saline conditions compared to both 5 mg/kg ($p = .032$) and 10 mg/kg ($p = .027$) drug dose groups. A

significant genotype * dose interaction was also reported ($F(3,44)= 3.30, p= .029$). Post hoc independent samples t-test comparing frequency of immobility between D₁ and wild type animals across 2.5 mg/kg, 5 mg/kg and 10 mg/kg drug dose groups were analysed in order to examine the nature of this interaction. A significant difference was found between D₁ and WT animals in the 5 mg/kg condition, $t(10)= 3.83, p=.003$, but not for 2.5 mg/kg or 10 mg/kg groups, See Figure 9.

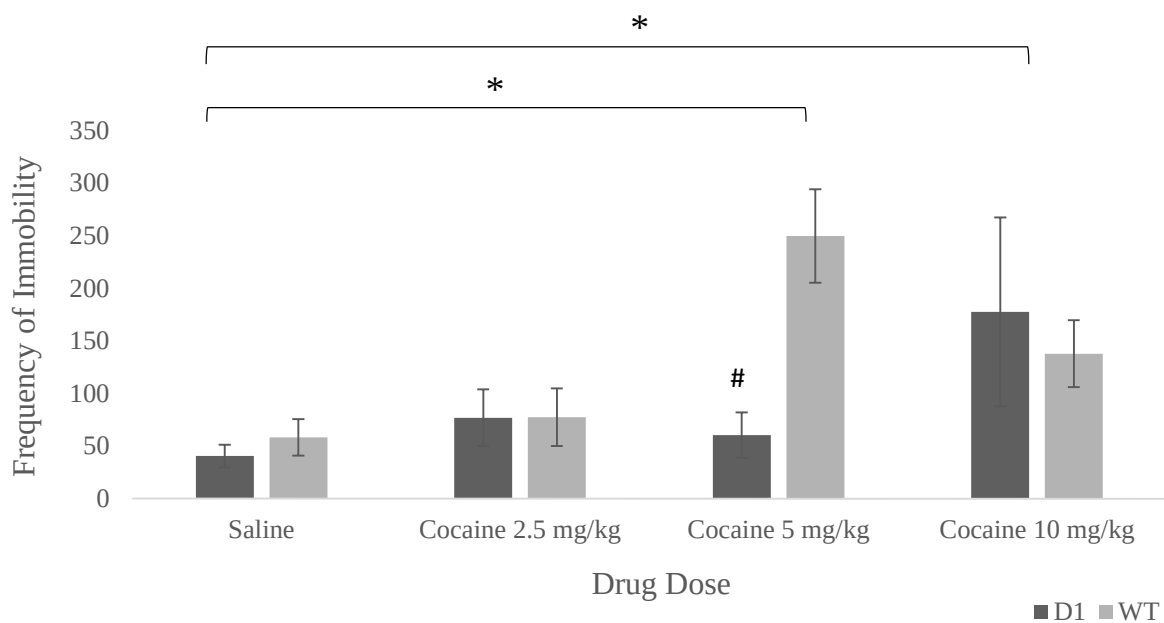


Figure 9. Frequency of immobility across cocaine drug doses. This graph clearly shows a large increase in start/stop behaviour among wild type animals under 5 mg/kg of cocaine. There are no significant differences between genotypes at other doses. Error bars represent SEM, * indicates significance at $p<.05$ level. # indicates a significant genotype difference ($p<.05$).

Rotation

Analysis returned no significant main effect of genotype or dose for either clockwise or counter clockwise rotation. No significant genotype*dose interaction was found. No significant genotype * dose interaction was found.

Head Weaving

No significant main effect of genotype or genotype * dose interaction was found for head weaving under cocaine conditions. However, a main effect of dose was discovered,

$F(3,44)= 11.20$, $p= <.001$. Post hoc Tukey tests were analysed to examine the nature of this effect. Results illustrated that there was a significant difference in the mean nose turning angle of animals administered saline compared to those administered both 5 mg/kg ($p= .028$) and 10 mg/kg ($p<.001$), indicating that head weaving behaviour is increased at lower doses of cocaine (see Figure 10). A significant difference was also reported between 2.5 mg/kg and 10 mg/kg dose groups, $p<.001$.

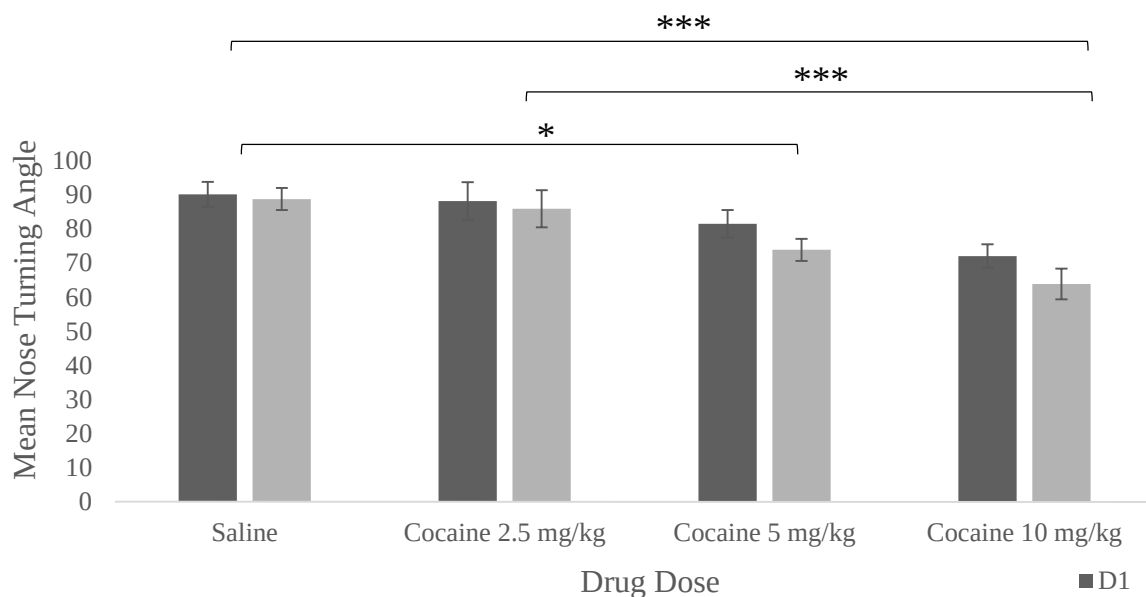


Figure 10. Mean nose turning angle in animals across cocaine drug conditions. This graph illustrates that overall wild types are exhibiting a slightly greater nose turn angle than D₁ animals. Head weaving behaviour also appears to decrease as drug dosage increases. Error bars represent SEM. ***, represents significance at $p<.001$, * indicates significance at $p<.05$ level.

Experiment Two: Conditioned Taste Avoidance

Methamphetamine

Methamphetamine CTA data was analysed using a 2(genotype: D₁, WT) x 3(dose: 0, 0.25, 0.5) ANOVA. A significant main effect of dose was revealed ($F(2,34)= 4.74$, $p= .015$) but not of genotype. No significant genotype * dose interaction was reported. Figure 11 illustrates that while methamphetamine-induced sucrose avoidance increased in a dose-dependent manner, there was no difference between genotypes. Post hoc Tukey tests showed that taste avoidance under 0.5 mg/kg conditions was significantly greater than under the

saline control condition ($p = .011$). For an outline of the descriptive statistics calculated for this experiment, please refer to Appendix A.

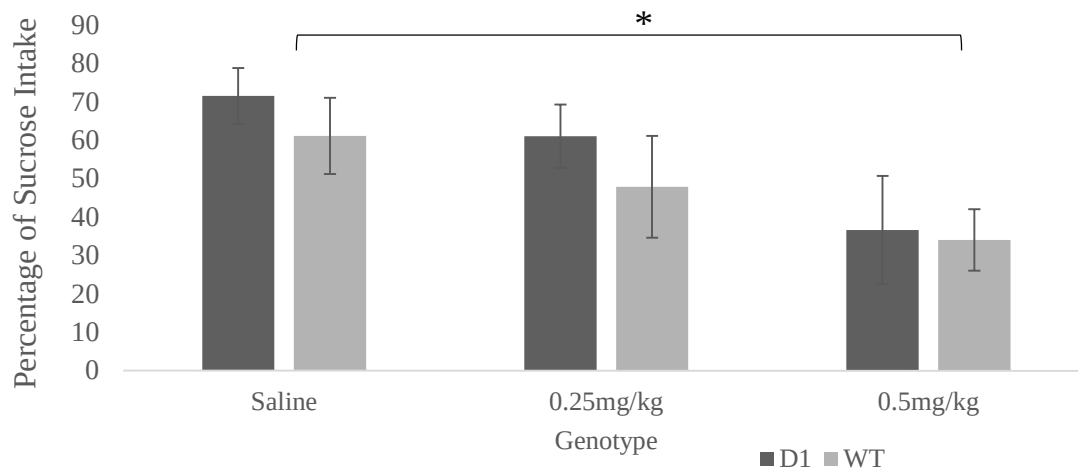


Figure 11. Methamphetamine CTA across drug doses and genotypes. This graph illustrates the degree of CTA (%) across saline and methamphetamine drug conditions for D₁ and wild type animals. Error bars represent standard SEM, * indicates significance at $p < .05$ level

Cocaine

A 2(genotype: D₁, WT) x 4(drug dose: 0, 2.5, 5, 10) ANOVA was carried out to examine the differences in cocaine-induced sucrose avoidance across genotype and dose conditions. No significant main effects of dose or genotype were reported, and no significant genotype * dose interaction was found. Results are also presented in Figure 11.



Figure 11. Cocaine CTA across drug doses and genotypes. This graph illustrates the degree of CTA (%) across saline and cocaine drug conditions for D₁ and wild type animals. Error bars represent standard error of the mean (SEM).

Experiment Three: Conditioned Place Preference

For an overview of the descriptive statistics, please refer to Appendix B. Paired samples t-tests were conducted to examine whether time spent in the drug paired compartment post-conditioning was significantly higher than pre-conditioning. Under methamphetamine conditions both D₁ ($t(15) = 4.36, p = .001$) and wild type animals ($t(15) = 2.54, p = .023$) showed significant CPP. The same was found for cocaine with D₁ ($t(15) = 4.58, p < .001$) and wild type ($t(15) = 2.97, p = .009$) results being comparable.

Methamphetamine ANOVA Results

A 2(genotype: D₁, WT) x 2(dose: methamphetamine 0.5 mg/kg, methamphetamine 1 mg/kg) ANOVA was conducted to compare the difference in preference increase for the drug-paired compartment post-conditioning between methamphetamine doses. No significant main effect of drug dose or genotype was found. However, analysis did reveal a significant genotype * dose interaction, $F(1,28) = 6.72, p = .015$ (see Figure 12.). Post hoc independent samples t-tests were carried out in order to examine the nature of this interaction. For the 1 mg/kg condition a significant difference between was found between D₁ and wild type rodents, $t(14) = 2.16, p = .049$, but not for the 0.5 mg/kg condition ($p = .150$). This indicates that, opposite to the pattern found under 0.5 mg/kg conditions, the D₁ animals showed a greater increase in preference for the drug-paired compartment than wild types, when administered 1 mg/kg methamphetamine doses during conditioning.

Cocaine ANOVA Results

A 2(genotype: D₁, WT) x 4(drug dose: cocaine 5 mg/kg, cocaine 10 mg/kg) ANOVA was carried out conducted to compare the difference in preference increase for the drug-paired compartment after conditioning between cocaine doses. No significant main effect of

dose or genotype was found as well as no significant genotype * dose interaction. Please refer to Figure 13.

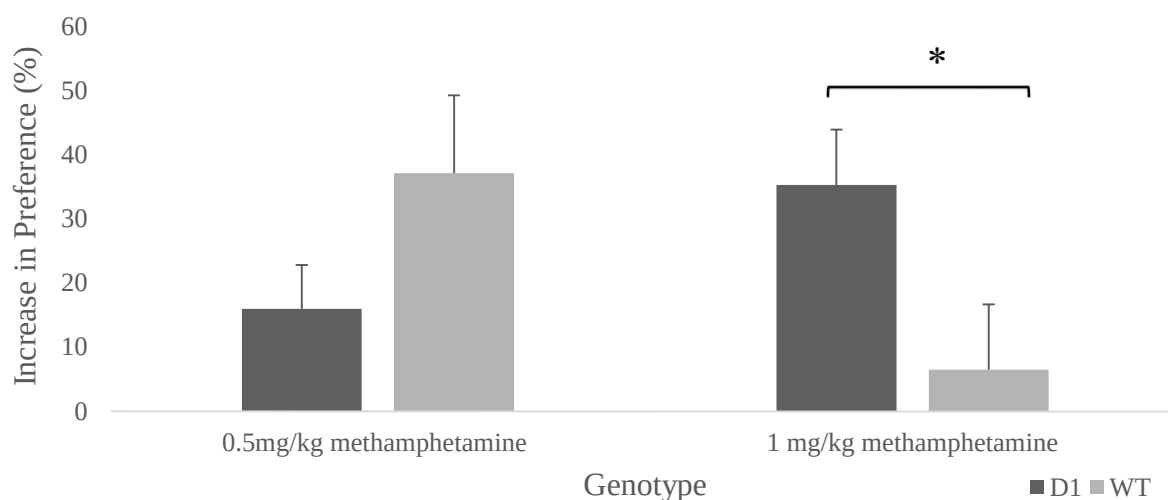


Figure 12. Increase in preference for the drug-paired chamber post conditioning. This graph illustrates the average increase in preference for the drug-paired compartment post-conditioning for both D₁ and wild type groups across methamphetamine drug doses. This figure clearly illustrates a drug dose*genotype interaction. The preference of wild type animals post-conditioning is much higher in the 0.5 mg/kg group as compared to the 1 mg/kg condition, whereas D₁ animals show the opposite pattern. Error bars represent SEM, * indicates significance at the $p < .05$ level.

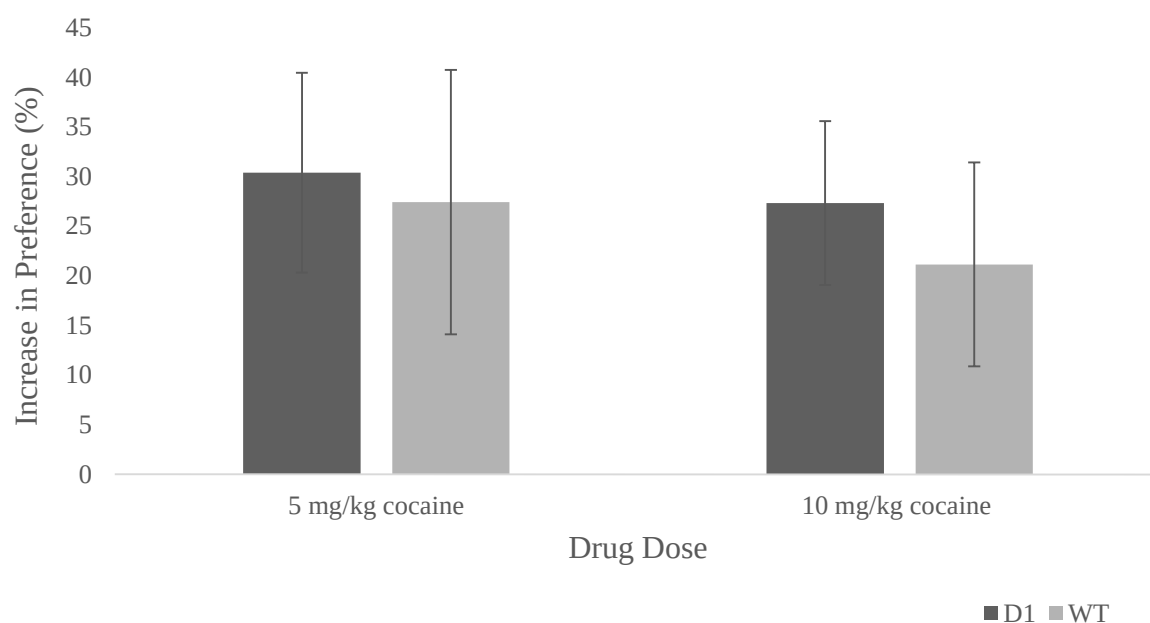


Figure 13. Increase in preference for the drug-paired chamber post-conditioning. This figure illustrates that after conditioning with either 5mg/kg or 10 mg/kg cocaine, both D₁ and wild type animals show comparable increases in place preference. It is clearly shown here that there are no genotype or dose differences occurring. Error bars represent SEM

Discussion

Summary of Findings

The aim of the current experiment was to further investigate the role of the dopamine D₁ receptor in the behavioural effects and rewarding properties of psychostimulants, specifically, in the effects of methamphetamine and cocaine. Three different experiments were conducted in order to examine this research question: a detailed analysis of locomotor activity, CTA, and CPP. Very briefly, for the locomotor experiment it was found that both cocaine and methamphetamine increased overall distance travelled in a dose dependent manner for both D₁ and wild type animals, as expected. More importantly, a significant genotype effect was found indicating that D₁ rodents travelled smaller distances across doses in comparison with wild type controls. In regards to the more detailed analysis that was undertaken it was found that methamphetamine had more specific genotype effects on a number of behavioural measurements (for example, immobility, rotation and head weaving). However, cocaine resulted in significant dose but not genotype effects. For a summary of the major findings please refer to Table 1.

For the CTA experiment, although the data appear to be trending in a dose dependent fashion, indicating that, as drug dose increases so does conditioned avoidance, statistical analysis only revealed a significant dose effect under methamphetamine conditions, finding a significant difference between avoidance at saline conditions and 0.5 mg/kg dose groups. No genotype effects were reported.

For our CPP experiments, findings illustrated that under cocaine conditions, no significant main effects of genotype or dose were found, and no significant interaction. The absence of a significant genotype effect contradicts the hypothesis that D₁ animals would show impaired expression of drug-induced place preference. Interestingly for

methamphetamine-induced CPP a significant genotype*dose interaction emerged which will be discussed below.

Locomotor activity

For a succinct summary of the major findings from our detailed locomotor analysis please refer to Table 1.

Table 1.

Summary of Locomotor Findings

Behavioural Measure	Methamphetamine			Cocaine		
	<u>G</u>	<u>D</u>	<u>G x D</u>	<u>G</u>	<u>D</u>	<u>G x D</u>
Distance travelled	X	X	-	X	X	-
Rearing	-	-	-	-	-	-
Immobility (duration)	X	X	-	-	-	-
Immobility (frequency)	X	X	-	-	X	X
Rotation (clockwise)	X	X	-	-	-	-
Rotation (counter clockwise)	X	X	-	-	-	-
Head Weaving	X	X	-	-	X	-

Results from the detailed locomotor analysis conducted in the current study fit in well with previous research. The fact that psychostimulant drugs such as cocaine and methamphetamine increase general locomotor activity is well established in the literature (Kita et al., 1999) and has again been confirmed here. Both control animals, with regular dopamine functioning, and D₁ mutants, with reduced expression of dopamine D₁ receptors brain wide, showed a dose-dependent increase in total distance travelled after both

methamphetamine, and cocaine administration. Although both genotypes showed increases in motor activity after drug administration, the D₁ rodents travelled significantly smaller distances than wild type controls, confirming our hypothesis. This finding supports previous research using antagonist drugs. For example, Kita et al. (1999) illustrated that pre-treatment with the selective D₁ antagonist SCH 23390 lead to a significant reduction in cocaine-induced motor activity. Localised administration of SCH 23390 into the medial pre-frontal cortex was also shown to diminish amphetamine- and methamphetamine-induced motor activation (Hall et al., 2009). The current findings support this previous work and confirm that a brain wide reduction in dopamine D₁ receptor functioning results in reduced cocaine- and methamphetamine-induced locomotor activity.

An important point to note here is the results from our saline control condition. No significant differences were found between genotypes for any of the locomotor behavioural measures analysed. This indicates that there is no difference in basal locomotor activity between these animals under normal circumstances. This is an important finding as alterations in the dopaminergic system often lead to altered motor activity, in line with dopamine's role in motor functioning (Wise, 2004). For instance, it has been found that movement disorders such as Parkinson's disease occur due to dopamine depletion in the striatum (Gerfen et al., 1990) and studies assessing the effect of D₁ antagonism (SCH 23390) on spontaneous motor activity in rats have revealed that disruption of D₁ receptor functioning leads to suppressed locomotor activity and rearing behaviours (Baldo, Sadeghian, Basso, & Kelley, 2002). Interestingly, research examining basal motor activity in D₁ mutant mice found that they exhibited hyperactivity (Xu et al., 1994). Receptor deficient mice were found to cause over double the amount of photobeam interruptions than wild type control mice. While this contrasts to the pharmacological findings that reduced D₁ functioning leads to decreased motor activity, it still supports the idea that D₁ receptors are important for the

regular expression of motor activity. These differences could be due to differences between rats and mice, or it could be a reflection of the fact that our mutant animals only show a 50% reduction in D₁ binding whereas these mice had a 95% deficit. Nonetheless, the current findings indicate that when not under drug influence motor activity between D₁ mutants and wild type controls is comparable and therefore will not be considered a confound in such drug-free experiments. Interestingly, as discussed in the introduction, previous research in our laboratory has shown that D₁ mutants rats are unable to learn to perform simple operant responses such as lever pressing and nose poking. In line of the above finding, it seems unlikely that this is due to a motor problem and thus seems more likely due to motivational deficits. This is also in line with other experiments from our lab showing that D1 mutant rats have deficits with reward perception and anticipation (Marie Vanden Broeke, unpublished thesis).

No effects of genotype or dose were found for either the duration of time spent in, or frequency of entries into, the rearing zone. This is unusual as dopamine is often believed to be involved in rearing behaviour and previous research examining amphetamine-induced locomotor activity have found significant increases in rearing behaviour (Wise, 2004). A possible reason why we did not find any significant effects for rearing is the way this parameter was measured in Ethovision. Typically, locomotor activity analysis is carried out using infrared beam methodology, which allows for easy measurement of rearing based on beam breakages. However, in Ethovision the animal is video recorded in an open field and this video is later analysed by creating specific arena setting and zones. These can sometimes be hard to define and the rearing zone can only be fixed around the outside edge of the open field, therefore, animals may be rearing in the middle of the arena and not being detected as entering the rearing zone, or they may not be rearing high enough. Additional manual scoring may shed more light on this

In addition to looking at the traditional parameters for assessing hyperactivity (i.e. distance travelled and rearing), we performed a much more detailed analysis, aimed at identifying more subtle changes between the genotypes. A greater number of parameters were analysed than presented within this report, but for the scope of this research we decided to focus on distance travelled, rearing, rotation, immobility and head weaving.

The immobility data indicate that under methamphetamine conditions D₁ animals exhibit longer durations of immobility but a smaller frequency of immobile periods in comparison with wild type controls. This means that when the mutant rats stopped moving they then remained motionless for longer periods than wild types. On the contrary, wild type rats exhibit an increased amount of start/stop behaviour, meaning they frequently stopped moving, but would quickly resume motor activity. A dose effect was also found illustrating that differences in immobility were most profound between 0.25 mg/kg and 0.5 mg/kg doses of methamphetamine. When the lower dose was administered the total duration of immobility for the testing session was much greater for both genotypes than it was at 0.5 mg/kg. This finding was not unexpected as after a low dose locomotor activity is only increased to a moderate degree implying that the rats still spent a substantial amount of time sitting still in the open field. Immobility increases again at the highest dose of methamphetamine resulting in a U shaped curve for wild type animals. This indicates that at higher doses behaviour becomes more constant and the start/stop behaviour that appeared under moderate dose conditions is diminished. For cocaine there was no effect of genotype or dose however, a significant interaction was found. This interaction indicates that specifically at 5 mg/kg wild type animals show a marked increase in frequency of immobility compared to D₁ animals. At other doses there was no difference between genotypes. These findings are similar to the pattern found in methamphetamine showing us that at low and high doses the frequency of stop behaviour is relatively low and comparable across genotypes. Presumably, this is

because at low doses animals are stopping for longer periods of time and at high doses motor activity becomes more constant. Previous research examining, in detail, the behavioural effects of *d*-amphetamine reported increases in freezing behaviour (Antoniou & Kafetzopoulos, 1991). It was found that at doses comparable to our study (1.0 – 1.5 mg/kg) freezing behaviour strongly featured in the behavioural profile of rats administered *d*-amphetamine. Rodents exhibited large frequencies of unpredictable stopping behaviour. This finding is in line with immobility results that revealed a large amount of start/stop behaviour in animals treated with methamphetamine.

Both clockwise and anti-clockwise rotation behaviour was examined. Under methamphetamine conditions a clear effect of genotype was found for both measures with wild type rodents exhibiting a marked increase in rotational behaviour compared to D₁ mutants. This behavioural profile was also effected by dose, rotation at the highest dose of 1 mg/kg was significantly increased compared to saline control conditions and also the lowest dose group of 0.25 mg/kg. For cocaine conditions no significant differences were found for rotational behaviour between genotypes or dose groups.

Head weaving behaviour was also analysed. In contrast to rotational data we found that the D₁ animals showed higher nose turning angles than wild type controls. On the surface this result appears surprising as if the wild type rats are showing more rotation and also general motor activation you would also expect them to exhibit large nose turning angles. However, when you examine this finding in conjunction with the general locomotor activity (distance travelled) data it makes more sense. Head weaving and general locomotor activity behaviours are to an extent, mutually exclusive. Increased head weaving would lead to a decrease in distance travelled, therefore it follows that if the wild type animals are travelling larger distances whilst inside the open field, they would also exhibit smaller nose turning angles. Based on this finding we posited that there should be a negative correlation found

between distance travelled and head weaving. Correlational analysis revealed that this was the case with these measures showing a strong negative relationship ($r = -.79$, $p < .001$). This idea is also supported by the fact that a significant effect of dose was found, indicating that nose turning angles are increased under saline and low dose conditions compared to moderate and high doses of methamphetamine. Interestingly, no differences in head weaving behaviour were found between genotypes under cocaine conditions. Similar to methamphetamine a dose effect is shown indicating that nose turning angles are increased at lower doses of cocaine. While the literature surrounding detailed locomotor analyses is limited, studies examining patterns of locomotor behaviours following cocaine and *d*-amphetamine administration have investigated head swinging behaviour (Antoniou & Kafetzopoulos, 1991, 1996; Antoniou et al., 1998). In contrast to our finding indicating that nose turning angles were increased at lower drug doses Antoniou and Kafetzopoulos (1991) reported that head swinging was induced by higher doses of amphetamine (6 mg/kg). This result was replicated by Antoniou et al. (1998), however, these results did not reach statistical significance. Cocaine was not found to produce any changes in head swinging behaviour (Antoniou & Kafetzopoulos, 1991; Antoniou et al., 1998). While head swinging and head weaving may seem like superficially related behaviours, it is possible that what the current study was measuring is different to the previously discussed literature. While it was not statistically significant, the increases in head swinging behaviour found after amphetamine administration most likely reflect stereotypy, whereas our increases in nose turning angle at low doses of methamphetamine and cocaine could be related to exploratory behaviour. Head weaving behaviour has also been strongly linked to serotonergic neural systems. Previous literature working with serotonin agonists and antagonists has shown a strong increase and decrease in head weaving behaviour, respectively (Lucki, Nobler, & Frazer, 1984; Nabeshima et al., 1984). Head weaving has been described as part of the serotonin behavioural syndrome (Tricklebank, Forler, & Fozard,

1984). The serotonergic nature of this behaviour may also contribute to the decrease in head weaving we saw at higher doses of both cocaine and methamphetamine in the current study. These drugs exert their effects primarily through dopamine release and at higher doses strong dopaminergic activation may overshadow the serotonergic actions producing this behaviour.

Overall the previous literature that has examined patterns of cocaine- and amphetamine-induced locomotor activity conclude that these two drugs lead to dissociable behavioural profiles. Cocaine has been shown to induce a behavioural profile characterised largely by increased motor activity and overall behavioural stimulation, whereas amphetamine-like drugs lead to a much more complex pattern of behaviour. High doses of amphetamine have been found to produce a large amount of behavioural switching and stereotypy (Antoniou & Kafetzopoulos, 1991, 1996; Antoniou et al., 1998). The findings from the current study do not appear to support this idea as both methamphetamine and cocaine were found to produce similar dose effects for a number of behavioural measures. However, it is possible that the dose-range we tested was simply too low to exhibit these previously reported findings. The previously discussed literature used doses up to six-times higher than our methamphetamine dose under amphetamine conditions and four-times higher for cocaine. This presents a clear question of why the current study did not use a higher dose range. The range used was chosen as we did not want to induce too strong an effect in the mutant animals that would have the potential to overshadow subtle genetic differences.

Overall, a clear pattern has emerged from our detailed locomotor analysis. It was found that under methamphetamine conditions significant genotype differences occur for a number of behavioural measures such as, distance travelled, immobility, rotation and head weaving. However, when animals are administered cocaine, genotype effects were only reported for distance travelled. This is an interesting finding and it provides support for our hypothesis that D₁ mutant animals would show an exaggerated deficit in responses when

administered methamphetamine as compared to cocaine. As discussed in the introduction both of these drugs are classed as psychostimulants and both exert their effects by working on the dopamine transporter. However, methamphetamine reverses the transporter where cocaine only blocks it. This means that methamphetamine administration results in a greater release of dopamine and therefore increased dopamine receptor stimulation. Because D_1 mutants have a down-regulated D_1 receptor system it follows that deficits would be more exaggerated following methamphetamine, compared to cocaine administration.

Conditioned Taste Avoidance

In the present study we found that while methamphetamine induced significant CTA, cocaine, in the dose range tested, did not. More importantly, in the present study we did not find a significant genotype effect or genotype * drug interaction. As discussed earlier, pharmacological research has shown that D_1 antagonist administration decreases the acquisition of lithium chloride-induced CTA (Fenu et al., 2001). However, the underlying processes of lithium chloride CTA are likely different to drug CTA which could explain the discrepancies between these and the current findings. Lithium chloride is an emetic agent that leads to gastrointestinal distress and produces strong aversive responses in rats assayed using the taste reactivity test. Drugs of abuse on the contrary are known to exert rewarding effects and have not been found to produce aversive taste reactivity.

Many theories have been proposed in recent years to try to explain why rewarding drugs of abuse induce CTA. The drug novelty hypothesis posits that any form of drug administration disrupts internal homeostasis in an animal and induces novel internal stimuli that they have not experienced before. These internal stimuli may be perceived as potentially dangerous to the animal and therefore will be avoided in future (Verendeev & Riley, 2012). This hypothesis has been criticised as it predicts that pre-exposure to a drug should subsequently decrease CTA, as the drug would not be novel anymore. However, while this

has been supported in one-trial conditioning, drug pre-exposure effects can be overridden with numerous conditioning trials which cannot be elegantly explained by the drug novelty hypothesis (Verendeev & Riley, 2012).

A more widely adopted theory explaining this paradox is the reward comparison hypothesis. Grigson (1997) put forward this hypothesis on the basis that the interpretation that rewarding drug-, and emetic drug-induced CTA both occur due to similar aversive properties between the drugs, is mistaken. Instead, it is suggested that the reason the drug-paired solution is avoided is due to the drugs reinforcing properties. This idea stemmed from the findings of Flaherty and Checke (1982) illustrating that rats would avoid a saccharin solution that had been paired with a highly rewarding sucrose solution. This was thought to illustrate that the animals learned the relationship between the saccharin and sucrose, and would avoid the first in anticipation of the more rewarding secondary solution. This was termed anticipatory contrast. Experiments showed that that rats would only avoid the saccharin if the sucrose was highly rewarding, as the 2% sucrose did not produce a significant avoidance but 32% did. Based on this it is proposed that drugs of abuse induce CTA as rats will avoid the conditioned solution (such as saccharin or sucrose) in anticipation of the much larger drug reward which will follow (Verendeev & Riley, 2012). This hypothesis therefore lends itself to the idea that drug-induced CTA relies, not only on the reinforcing properties of the drug of abuse that is administered, but also of the solution they are paired with. According to this hypothesis if the sucrose or saccharin solution is considered more rewarding than the administered drug to the animal, then it will not be avoided.

While investigation into the role of the D₁ receptor in drug-induced CTA has been limited in the literature, Hunt et al. (1985) illustrated that pimozide decreases cocaine-induced saccharin avoidance, this finding is restricted in its comparability to our work due to the selectivity of pimozide. Pimozide is a non-selective dopamine antagonist that works on

D₃ and D₄ receptors as well as some serotonin receptors; however, pimozide has a particularly high affinity for dopamine D₂ receptors. Therefore, as other work examining the role of the D₁ receptor in cocaine CTA does not appear to have been carried out, it is possible that receptors from the D₂ family are mediating this effect found by Hunt et al. (1985) and D₁ receptors may not be strongly involved in the expression cocaine-induced taste avoidance. For the current experiment, we tested avoidance to a five percent sucrose solution conditioned through pairings of either saline, 2.5 mg/kg, 5 mg/kg or 10 mg/kg cocaine. A limitation with this data set is the amount of variability and the relatively small group size. Apart from the saline condition, which had eight animals per group, each drug condition only has a sample of six D₁ and six wild type rodents. We would have liked to run more however breeding problems meant that unfortunately, there were no other animals available. It is also possible that even after adding the 10 mg/kg cocaine dose group the dose range was too small to find an effect. The doses we used were chosen so a consistent dose range could be tested between all experiments. However, previous research that found significant dose-dependent cocaine-induced CTA used a dose range of 10 – 36 mg/kg (Goudie, Dickins, & Thornton, 1978). Future studies should therefore try to replicate this experiment using a larger N and higher dose range.

The fact that no significant genotype effects were found for the current experiment supports our hypothesis that no differences would be observed in taste avoidance between genotypes after methamphetamine. Moreover, it extends these findings to cocaine. Referring back to the previous literature work carried out looking at the role of the D₁ receptor in amphetamine-induced CTA by Lin et al. (1994) found that only if both SCH 23390 and raclopride were given together as a pre-treatment then *d*-amphetamine-induced saccharin avoidance was decreased. However, if SCH 23390 or raclopride were administered alone no significant effect was found. This provides support for the argument that neither D₁ nor D₂

receptors alone are mediating CTA and instead both families of receptors play an important role. Our data fit in nicely with this literature as while D₁ mutants have deficits in D₁ receptor functioning, other dopamine receptors are left unchanged. If concurrent activation of both the D₁ and D₂ family of receptors is important for mediating drug-induced CTA, then it follows that the D₁ mutant rats would show little deficit in the expression of this behaviour.

A possible confound that also needs to be discussed in regards to our CTA experiment is the solution we used. While most previous literature examining both taste aversion and avoidance uses saccharin, we used five percent sucrose. As discussed earlier the reward comparison hypothesis argues that the avoidance induced by drugs of abuse depends, not only on the rewarding properties of the drug, but also of the gustatory conditioned stimulus (Verendeev & Riley, 2012). As the hypothesis posits that animals avoid the rewarding solution in anticipation of a more rewarding drug delivery, it therefore follows that the solution must be less rewarding to the animal than the drug, in order for taste aversion to occur. It is possible that for our experiment the animals did not value the sucrose solution as markedly less rewarding than the doses of methamphetamine or cocaine that were administered. The only major difference between saccharin and sucrose is the calorific value, being that saccharin contains no calories. However, saccharin is often described as having a bitter after taste and has been shown to be less favoured by rats. A study conducted by Self (2010) examined the preference of a wide range of sucrose and saccharin doses. It was found that at low doses of both saccharin and sucrose (0.2-0.4% and 2-4%, respectively) preference across solutions was equal. However, at increasing concentrations preference for saccharin decreases considerably. As our sucrose solution was made at a concentration of five percent it could be argued that the reward value was too high. Based on this it would be recommended that future CTA experiments in our laboratory be carried out using saccharin rather than sucrose solutions.

Conditioned Place Preference

In the CPP paradigm, we found a significant gene * methamphetamine interaction but no effect of cocaine.

Our CPP findings illustrate that D₁ animals are not impaired in acquisition of cocaine-induced place preference, but they do show a deficit under methamphetamine conditions. No significant genotype effect was found for cocaine, which indicates that increase in preference for the drug-paired chamber post conditioning was comparable across genotypes. However, a genotype * dose interaction was found for methamphetamine indicating reduced place preference sensitivity compared to wild type animals. This finding partially supports our hypothesis, which stated that the D₁ mutant animals would show diminished CPP expression compared to wild type control animals. Our hypothesis was based on an extensive review of previous research that has examined the effects of dopamine antagonists on drug-induced CPP. Graham et al. (2007) administered drug naïve rats with three different D₁ receptor agonists (ABT-431, SKF 81297, and SKF 82958) and all were shown to dose-dependently induce CPP, however when D₂ agonists were administered no effect was found. This work indicates that D₁ receptors, and not D₂, are an important target needed for the acquisition of CPP. Perhaps more relevant, work using antagonist drugs has also previously found that both SCH 23390 and SCH 39166 eliminate CPP induced through amphetamine administration (Acquas & Di Chiara, 1994). Hiroi and White (1991) even found that D₁ receptor antagonism successfully blocked the expression of amphetamine CPP both when given prior to conditioning, and also when administered systemically after conditioning. Work done on cocaine-induced CPP has shown similar findings with many authors reporting that D₁ antagonist administration blocks CPP (Baker et al., 1998; Cervo & Samanin, 1995; Arbi Nazarian, Scott J Russo, Eugene D Festa, Mohammed Kraish, & Vanya Quinones-Jenab, 2004). Taken together these findings provide strong evidence for the idea that D₁ receptors

are essential for the acquisition and expression of CPP. This makes the current results found under cocaine conditions surprising. On the other hand, work using genetic animal models with disruptions in their dopamine systems have examined the importance of the D1 receptor in cocaine-induced CPP and found that it can still be established. For example, D₁ knockout mice were found to exhibit place preference for the cocaine-paired environment at a level comparable to wild type controls (Miner, Drago, Chamberlain, Donovan, & Uhl, 1995), in line with our own finding using D1 mutant rats. The results from pharmacological and genetic model research on cocaine CPP, pose an interesting contradiction within the literature. Possible explanations for this discrepancy could include a dosing effect, or neural compensation. These ideas will be discussed at length shortly as they also relate to our CTA findings and should be considered as a general confound for the current study.

A very interesting finding from the current CPP experiment was the significant genotype * dose interaction found for methamphetamine. This result illustrated that when administered a moderate dose of methamphetamine (0.5 mg/kg) wild type animals found this reinforcing and exhibited a significant increase in preference for the drug-paired chamber. Comparatively the D₁ animals did not exhibit strong CPP at this dose. However, when exposed to a higher dose of methamphetamine during conditioning (1 mg/kg) wild type animals showed the opposite pattern, illustrating no significant change in preference for the drug-paired compartment. While the results were not strong enough to actually elicit a significant place aversion (defined as a significant increase for the saline paired chamber) the findings indicating that this higher dose of drug may have been starting to be experienced as more aversive (indeed several animals showed an increase preference for the saline paired compartment). D₁ mutant animals, in contrast, showed very convincing preference for the drug-paired compartment, to a higher degree than when tested under 0.5 mg/kg conditions, illustrating that this was an optimal dose level for D₁ animals. This finding, similar to our

locomotor results, provides support for the idea that dopamine D₁ receptor mutant animals have a reduction, rather than a complete eradication of the membrane expression of the D₁ receptor. At the lower dose of methamphetamine, mutant rats show a decreased conditioned preference, this can likely be attributed to reduced dopamine stimulation, which subsequently results in reduced drug reinforcement. In contrast, when the higher dose of 1 mg/kg was administered D₁ animals expressed a large increased in preference for the drug-paired compartment, indicating that a larger increase in extracellular dopamine is necessary for optimal stimulation of dopamine D₁ receptors in these animals. For the wild type rats, this high dose condition did not produce place preference, with many animals favouring the saline-paired compartment. This indicates that 1 mg/kg of methamphetamine was too high a dose and may have created an adverse effect rather than acting as a reinforcer. These findings represent a shift to the right of the dose response curve for D₁ mutants and support previously discussed pharmacological findings illustrating that D₁ antagonism leads to a reduction in the expression of amphetamine-induced CPP.

General Considerations

Taken together the current findings provide some interesting insight into the role of the D₁ receptor in the behavioural and reinforcing properties of cocaine and methamphetamine. However, before this is discussed some general considerations or limitations of our experiments should be addressed.

A limitation of the present CPP findings is the lack of a saline control condition. Due to animal constraints it was not possible to include groups of wild type and D₁ mutant animals in this study that received saline in both compartments. This would have allowed us to control for change in preference scores that were simply due to repeated exposure to the

chamber. Replication of our CPP findings in future experiments incorporating this control would be useful to aid interpretation of these results.

An important aspect of working with genetically altered animal models is the presence of neural adaptation (Self, 2010). It is possible that as a result of this missense mutation in the D₁ receptor, developmental compensation has occurred in the brains of these rodents leading to up regulation of other receptor systems. This could explain why for both out CTA and CPP studies (cocaine condition only) D₁ animals did not exhibit a decreased sensitivity to the reinforcing properties of methamphetamine and cocaine. However, previous work carried out on this mutant animal model has examined alternate dopamine receptor systems and no significant compensatory upregulations were found. For instance, the mRNA levels of the remaining four dopamine receptors was unaltered (unpublished data). The fact that the mutation leads to only a 50% reduction in the membrane expression of the D₁ receptor, rather than a complete eradication, is also thought to reduce neural adaptation.

Another factor at play that could be affecting our findings is the localisation of the mutation. The D₁ mutation is brain wide, meaning that the functioning of dopamine D₁ receptors has been critically altered throughout all neural circuits and regions. Out of the battery of experiments we conducted in the current study, the role of dopamine in locomotor activity is arguably more straightforward than its role in the reinforcing properties of drugs. This may explain why we found such a clear decrease in sensitivity to both methamphetamine- and cocaine-induced locomotor activity in D₁ animals, but no significant differences between genotype performance in the CTA and cocaine CPP paradigms. When researching drug addiction and the rewarding properties of psychostimulants, the most implicated brain region is the mesocorticolimbic dopamine system. Specifically, neuron projections from the ventral tegmental area to the nucleus accumbens. A recent review conducted by Chaudhury, Liu, and Han (2015) describes the idea that there are different

groups of dopamine cells within the ventral tegmental area that are functionally distinct. It is believed that these different neurons are involved in different projection circuits and they encode disparate behaviours. For example, activation of inputs from the laterodorsal tegmentum to the ventral tegmental area stimulates reward behaviours, whereas inputs from the lateral habenula elicit aversive behaviours. It may be that dopamine D₁ receptors localised in very specific neural regions play an important role in the rewarding effects of cocaine and methamphetamine. Therefore, the D₁ mutant rodent, due to having brain wide deficits, as well as systemic pharmacological studies which lead to wide spread receptor blockade, may not be sensitive enough models to pick up these effects. Advances in genetic deletion strategies in mice, have led to the development of inducible localised genetic deletions. This means that researchers have the ability to breed animals with regular brain functioning that can be altered at any time within the animals' life span. This is done by using binary transgenic systems where the expression of a target gene becomes controlled by interaction between an effector and target transgene (Lewandoski, 2001). Moreover, techniques are available for regional-selective (rather than global) genetic alterations. Optogenetics is another popular technique used for precise gene targeting. Through genetic manipulation, specific (sub)types of cells are altered to make them selectively respond (turned on or off) using a light stimulus. This gives researchers very detailed control over selected cells and specific cellular events (Deisseroth, 2011). Future research assessing the specific role of dopamine receptors would benefit from the introduction of a rat models utilizing these techniques. This would allow for close examination of the role of the D₁ receptor in specific areas throughout the brain. Also the ability to induce gene deletion at any stage means that the risk of neural compensation is eradicated, and animals do not suffer from any neural deficits during sensitive developmental periods.

An interesting addition to future projects of this nature would be to run an extra group of animals. Running wild type animals that were pre-treated with the D₁ antagonist SCH 23390 would be a valuable additional measure, and help aid in interpretation of our findings. While, as we have previously mentioned, it is hard to interpret antagonist effects due to the non-selective nature of these drugs, it would be interesting to observe any differences that might arise between D₁ mutant rodents, who have a reduced expression of D₁ receptors from birth, and wild type animals experiencing temporarily reduced D₁ functioning as a result of antagonist administration.

While this work is largely preliminary in nature, research examining the role of specific dopamine receptor systems is important. The more advances we make in understanding the role of dopamine and its' receptor families in drug reinforcement, the closer we get to being able to develop successful target-based pharmacological treatments for drug dependence. While the findings from the current study imply that D₁ receptors alone are not playing a major role in the reinforcing properties of methamphetamine and cocaine as assayed by CTA and CPP paradigms, they do illustrate an interesting difference between methamphetamine and cocaine. A major finding that has emerged from this research is that D₁ animals show more exaggerated deficits or differences in responding compared to wild types under methamphetamine rather than cocaine conditions. This result supports our original hypothesis and implies that D₁ receptor stimulation is enhanced during methamphetamine administration compared to cocaine, even though cocaine and methamphetamine are thought to work in very similar ways. The fact that D₁ animals showed significant, but not exaggerated deficits in locomotor behaviours throughout this experiment also supports the claim that the genetic mutation found in these animals leads to reduced functionality of the D₁ receptor system. Previously, using a [³H]SCH23390 label, dopamine D₁ receptor binding was examined in ex vivo tissue of these mutant rodents. Autoradiography

results revealed that D₁ binding was decreased by up to 50% in different brain regions of these animals (unpublished data). Our locomotor and methamphetamine CPP findings, and the observed shift to the right of the dose response curve for many of the assessed behavioural measures, confirms this finding.

In conclusion, while our findings do not directly implicate the dopamine D₁ receptor in the reinforcing properties of cocaine and methamphetamine they have revealed a potential dissociation in D₁ receptor functioning between these drugs. While it is possible that the effects we have found here are due to a low dose range used under cocaine conditions, based on the large amount of previous literature indicating differences in the mechanistic profiles of dopamine in cocaine and methamphetamine this is not believed to be the case. However, work using higher dose ranges is recommended in the future. Due to the amount of contradiction between pharmacological antagonist and genetic model findings within the literature surrounding CPP and also our CTA findings, more research directly comparing these approaches could also be beneficial. Also, research examining cocaine and methamphetamine reward using different paradigms is necessary. While previous attempts to get D₁ animals to lever press have been unsuccessful, assessing the performance of these animals in a self-administration study, if possible, would be the logical next step. As the current study has again confirmed that the D₁ mutants do not show motor deficits it is conceivable that, if a larger amount of training, or perhaps more pre-exposure to drug occurred, these animals could learn to press a lever or perform a similar operant response in order to receive drug delivery. If possible, this would extend the current findings and help develop the current knowledge of the role of D₁ dopamine receptors in drug reinforcement.

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Appendix A

Conditioned Taste Avoidance Descriptive Statistics

Table A1.

*Conditioned Taste Avoidance Descriptive Statistics*D₁ Mutant Animals

<u>Drug Dose</u>	<u>Mean CTA (%)</u>	<u>Standard Deviation</u>
Saline	71.51	20.70
Methamphetamine 0.25mg/kg	61.00	20.20
Methamphetamine 0.5mg/kg	36.58	34.61
Cocaine 2.5mg/kg	61.68	39.70
Cocaine 5mg/kg	53.73	25.08
Cocaine 10mg/kg	48.54	37.29

Wild-Type Animals

<u>Drug Dose</u>	<u>Mean CTA (%)</u>	<u>Standard Deviation</u>
Saline	61.11	28.00
Methamphetamine 0.25mg/kg	47.82	32.47
Methamphetamine 0.5mg/kg	34.02	19.65
Cocaine 2.5mg/kg	67.06	20.50
Cocaine 5mg/kg	66.69	25.92
Cocaine 10mg/kg	40.04	14.85

Appendix B

Conditioned Place Preference Descriptive Statistics

Table A2.

*Conditioned Place Preference Descriptive Statistics*D₁ Mutant Animals

<u>Drug Dose</u>	<u>Mean Time Pre-conditioning</u>	<u>Standard Dev.</u>	<u>Mean Time Post-conditioning</u>	<u>Standard Dev.</u>	<u>Mean Increase in Preference</u>	<u>Standard Dev.</u>
Methamphetamine 0.5mg/kg	4.23	1.46	6.62	4.02	15.97	19.40
Methamphetamine 1mg/kg	3.95	1.41	9.24	3.13	35.28	24.41
Cocaine 5mg/kg	4.39	2.19	8.94	4.11	30.37	28.41
Cocaine 10mg/kg	4.52	2.26	8.62	2.92	27.32	23.36

Wild Type Animals

<u>Drug Dose</u>	<u>Mean Time Pre-conditioning</u>	<u>Standard Dev.</u>	<u>Mean Time Post-conditioning</u>	<u>Standard Dev.</u>	<u>Mean Increase in Preference</u>	<u>Standard Dev.</u>
Methamphetamine 0.5mg/kg	3.82	1.61	9.40	4.23	37.16	34.27
Methamphetamine 1mg/kg	4.86	1.38	5.84	3.86	6.48	28.75
Cocaine 5mg/kg	3.79	2.83	7.90	5.23	27.41	37.31
Cocaine 10mg/kg	4.69	2.27	7.86	4.65	21.13	29.08